



Performance evaluation of due-date based dispatching rules in dynamic scheduling of diffusion furnace

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Abstract

There has been extensive scheduling research relating to use of existing dispatching rules along with/without new dispatching rules and compared their performance behavior in job-shop, flow-shop, open-shop, flexible manufacturing system, and single machine with unit capacity environments using various scheduling objectives. However, it appears that there is no comparative study on analysis of dispatching rules for scheduling bottleneck batch processing machine in discrete parts manufacturing, particularly the diffusion furnace (DF) in semiconductor manufacturing (SM). This study addresses this research issue. For that, this study first, proposes the mathematical models for dynamic scheduling (DS) of DF to optimize the due-date based scheduling objectives: Total weighted tardiness, on-time delivery rate, total earliness/lateness, and maximum lateness. Due to the computational intractability of each the proposed mathematical models for large-scale problem, this study proposes greedy heuristic algorithm (GHA) based on due-date based dispatching rules (DDR). Because, dispatching rules are widely used in the SM industry. Accordingly, in this study twenty variants of GHA-DDR are proposed by considering various due-date based dispatching rules to compare the effects of due-date based dispatching rules in DS of DF. From the series of computational analysis carried out in this study, it is observed empirically that the proposed variants of GHA based on apparent tardiness cost (ATC) and batch ATC (BATC) dispatching rules yield consistently better solution for most of the scheduling objectives considered in this study. This observation is further verified by statistical analysis: Friedman test and Nemenyi multiple comparison test.

Keywords Batch processing machine · Greedy heuristic algorithm · Dynamic scheduling · Due-date based dispatching rule · Due-date based scheduling objectives

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1 Introduction

This research concerns on scheduling of a BPM particularly a diffusion furnace (DF) in semiconductor manufacturing industry. The efficient scheduling of DF is particularly important as processing time for diffusion operation is extremely large (10 h) when compared to other operations (1 to 2 h) in the wafer fabrication area [40] of semiconductor manufacturing industry. Among various methodologies available in the literature to address scheduling of DF, dispatching rules are widely used in this industry [19] as they (a) are perform reasonably well in a wide range of environments, (b) are relatively easy to implement and understand, (c) need only minimal computational time, and (d) can manage dynamic changes.

In general, dispatching rules are broadly classified into (1) simple dispatching rule, and (2) composite dispatching rule. A composite dispatching rule is a ranking expression that combines a number of simple dispatching rules. The simple rule is a function of a few attributes of the jobs (e.g., priority, weighted processing time, due-date, etc.) and/or the machines (e.g., speed, the number of jobs waiting for processing, etc.). An attribute of the job and/or machine may be either constant or time dependent. Based on that, the dispatching rules are classified as static or dynamic rule. In static rule, all the attributes are constant. Whereas the dynamic rule has the time dependent attribute(s) [45].

Though there has been extensive scheduling research relating to use of existing dispatching rules along with or without new dispatching rules and compared their performance behavior in job-shop, flow-shop, open-shop, flexible manufacturing system, and single machine with unit capacity environments using various scheduling objectives, there is no such analysis of due-date based dispatching rules (DDR) for dynamic scheduling (DS) of bottleneck batch processing machine in general, particularly the diffusion furnace (DF) in Semiconductor Manufacturing. This research issue is addressed in this paper by proposing the twenty variants of greedy heuristic algorithm (GHA) based on various DDR (GHA-DDR) to optimize the due-date based scheduling objectives: total weighted tardiness (TWT), on-time delivery (OTD) rate, total earliness/lateness (TE/L), and maximum lateness (Lmax).

The paper is organized as follows: the problem statement with assumptions is presented in Sect. 2. The closely related literature review is presented in Sect. 3. The proposed mathematical model and its workability are discussed in Sect. 4. Proposed greedy heuristic algorithm (GHA) based on ‘due-date based dispatching rules’ (DDR) for dynamic scheduling of diffusion furnace is discussed in Sect. 5. Performance evaluation of the proposed twenty variants of GHA-DDR for DS-DF is analyzed both (1) empirically in comparison with optimal solution as well as estimated optimal solution and (2) statistically in Sect. 6. Finally, a summary of contributions along with possible future research directions are discussed in Sect. 7.

2 Problem statement and assumptions

There are ‘N’ jobs representing ‘F’ incompatible job-families. And each job, ‘j’ of a family f ($f = 1, 2 \dots F$) has a processing requirement of PT_f , a due-date of DD_j , and a non-zero unequal release time of RT_j (at which it becomes available for processing), and the release time and due-dates are non-agreeable (that is, if $RT_i \leq RT_j$ not-implied $DD_i \leq DD_j$). Further, the processing time of every job in the same family is equal and jobs from different families cannot be processed in the same batch due to diffusion operation’s chemical nature in the wafer fabrication. The ready time of a batch is given by the longest ready time of all jobs in the batch. With these given parameters and constraint, this study optimizes the due-date based scheduling objectives: TWT, OTD rate, TE/L, and Lmax, considering one at a time as a single objective in dynamic scheduling of DF.

Total weighted tardiness (TWT) is vital to avoid important jobs to be tardy (late). Tardiness of a job ‘j’ (T_j) can be defined as how much late the job ‘j’ is processed as compared to the due-date (i.e.) positive lateness [$T_j = \max(0, CT_j - DD_j)$, where CT_j and DD_j are completion time and due-date of job ‘j’, respectively]. The weighted tardiness of the job ‘j’ (WT_j) can be defined as

$$WT_j = W_j \times T_j$$

where W_j is weight (that is priority or value or simply important) of the job ‘j’.

On-time delivery (OTD) rate is aiming to complete maximum number of jobs on or before its due-date. It can be defined as out of ‘N’ jobs how many jobs are completed on or before its due-date. It can be defined as follows

$$OTD \text{ rate} = \sum_{j=1}^N OTD_j / N$$

where $OTD_j = 1$ if $CT_j \leq DD_j$; 0 Otherwise.

Total earliness/lateness (TE/L) is sum of the absolute deviations of completion times from the due-date of all jobs. In practice, it may occur when the job ‘j’ is completed before its due-date (DD_j) then earliness penalty is incurred. So completing the jobs too early would also create problem. Hence, minimizing TE/L makes sure to avoid completing the jobs too early and too late. It can be defined as follows

$$TE/L = \sum_{j=1}^N |CT_j - DD_j|$$

$L_{\max}$ is the maximum lateness among all jobs. Minimizing the maximum lateness is in a sense equivalent to minimizing the worst performance of the schedule. It can be defined as follows

$$L_{\max} = \max_{j=1,2,\dots,N} (L_j)$$

where L_j is lateness of the job ‘j’.

To concisely describe the problem, we shall use the three-field notation of Graham et al. [17] to denote the scheduling of single DF problem as “1/batch, dynamic job-arrivals, incompatible job-families, non-agreeable release times and due-dates/TWT, OTD rate, TE/L, Lmax” and we make the following assumptions:

- All data related to dynamic scheduling of a DF problem considered in this study are assumed to be deterministic and known a priori. In practice, estimates of the required parameters’ values for this problem can be obtained from the existing shop-floor computerized information system, which tracks work-in-process inventory and the machine status in real time.
- Each job requires one operation and all jobs are independent.
- The DF has a capacity ‘B’, measured in terms of number of jobs it can hold. The number of jobs in a batch must be less than or equal to ‘B’. All jobs in a batch must be of the same family.
- Once processing of a batch is initiated, it cannot be interrupted and other jobs cannot be introduced into the machine until processing is completed.

3 A closely related literature review

Dispatching rules have been studied extensively in the literature. Panwalkar and Iskander [44] provided a good analysis of dispatching rules in scheduling. As this study considers only due-date based scheduling objectives, the earlier research on due-date based dispatching rules (DDR) to optimize various due-date based scheduling objectives are reviewed. It is observed that, over the past few decades, there has been extensive research relating to use of existing dispatching rules along with or without new dispatching rules and compared their performance behavior in job shop [51], flow shop [14], open shop [20], flexible manufacturing system [27], and single machine with unit capacity environments [8] for various due-date based scheduling objectives in various industries.

Further, it appears from the literature that, there are some studies [34, 48, 50] used the existing dispatching rules along with or without new dispatching rules and compared their performance behavior in scheduling problems related to wafer fabrication in semiconductor manufacturing industry. In the wafer fabrication process, one of the important operations is diffusion process, as diffusion operation requires relatively longest processing time (10 h) among all the operations in the wafer fabrication. In addition to that, around 30% of the total WIP (work in process) in a wafer fabrication is accumulated in the diffusion stage [24]. Due to these, diffusion operation is the bottleneck operations in the wafer fabrication. However, our literature review clearly indicated that, there is no comparative study on analysis of various due-date based dispatching rules in scheduling of diffusion furnace [a batch processing machine (BPM)] in semiconductor manufacturing (SM) industry.

The problem of scheduling a BPM in various discrete parts manufacturing industries have been addressed extensively in the literature (Table 1). Furthermore, Mathirajan and Sivakumar [35] reported that many research studies have been carried out on scheduling BPM in SM from deterministic perspectives.

Table 1 Sample references on scheduling BPM research in discrete parts manufacturing

Industry/manufacturer	Operations/BPM	Sample references
Aircraft industry	Hardening of synthetic parts/oven	[32]
Automobile gear manufacturing industry	Hardening and soaking/heat-treatment furnace	[47]
Clothing industry #	Vat (dyeing machine)	[58]
Electronics manufacturing industry	Testing PCBs/environmental stress screening (ESS) chambers	[9]
Furniture manufacturing industry	Dry Kiln	[57]
Glass container industry	Annealing/Lehr or annealing Kiln	[11]
Hospital #	Washer	[43]
Ion plating (IP) industry	IP machine (has been widely used in watch and clock industry)	[3]
Iron and steel industry	Multi-head hole-punching machine	[42]
Multi-layer-ceramic capacitor manufacturing industry	Bake-out/box-oven	[26]
Semiconductor manufacturing industry	Oxidation or diffusion furnace, burn-in oven	[23]
Shoe manufacturing industry	Carousel rotation/carousel	[12]
Steel casting industry	Heat-treatment furnace	[36]
Steel ingot production industry	Preheating/soaking bit	[16]
Thin film transistor liquid crystal industry	Aging test/chamber	[4]

Service industry is also included in Table 1 to complete the lists of BPM research, discussed in literature

Since the entire shop-floor is computerized for tracking purpose in SM, the deterministic assumption is widely accepted by research communities. Therefore, we also focus the deterministic scheduling of DF in this study. The deterministic scheduling of DF can be classified into static and dynamic. A static scheduling of DF assumes that all the jobs are available at time zero. Whereas dynamic scheduling of DF assumes that, each job has different non-zero release time (at which the job becomes available for processing). This study focuses on dynamic scheduling (DS) of DF.

In the literature, the deterministic and dynamic scheduling of DF deals with various scheduling objectives. It can be broadly classified as (a) completion time based objectives (Makespan, total flow time, mean flow time, etc.), and (b) due-date based objectives [$L_{\max}$, Number of tardy jobs, total tardiness (TT), Total weighted tardiness, etc.]. As the due-date based objectives are more important to deliver the goods to the customer on time in order to enhance the customer's satisfaction and strengthen competitive advantage [50], this study concern about due-date based scheduling objectives.

As this study considers due-date based scheduling objective, the available literature on the deterministic and dynamic scheduling of DF deals with various due-date based scheduling objectives, considering/proposing various due-date based dispatching rules (DDR) such as earliest due-date (EDD), flow due-date (FDD), operational due-date (ODD), modified operational due-date (MOD), critical ratio (CR), minimum slack (MS) First, Cost OVER time (COVERT), and apparent tardiness cost (ATC) and/or batch apparent tardiness cost (BATC) are presented in Table 2. From Table 2, it is observed that, (1) though many researchers used/proposed different due-date based dispatching rules for scheduling DF, there is no study which explicitly compares all possible due-date based dispatching rules in scheduling DF to arrive efficient due-date based dispatching rules, (2) there is no study consider the real-life requirements on 'non-agreeable property' of scheduling DF, except the study by Vimala Rani and Mathirajan [54], Vimala Rani and Mathirajan [55], and Vimala Rani and Mathirajan [56], and (3) there is no study consider the OTD rate and TE/L as the scheduling objective, except the study by Vimala Rani and Mathirajan [55]. This study attempts to fill all these gaps in dynamic scheduling of DF problem.

4 Proposed mathematical models

The nomenclature and the formulation of the proposed mathematical models to optimize TWT, OTD rate, TE/L, and $L_{\max}$ are discussed in this section.

A nomenclature

Sets:

J Jobs

F Families

Table 2 A summary of deterministic and dynamic scheduling of DF research with due-date based scheduling objectives considering various DDR

References	Number of DF		Release time and due-date considered as		Due-date based dispatching rules (DDR)							Due-date based scheduling objective		
	Single	Multiple	Agreeable	Non agreeable	EDD	FDD	ODD	MOD	CR	MS	COVERT		ATC and/or BATC	
Uzsoy [52]	✓		✓		✓									Lmax/Cmax/TWCT
Monch et al. [38]		✓	✓									✓		TWT
Dirk and Monch [10]		✓	✓									✓		TWT/Cmax
Monch et al. [40]		✓	✓									✓		TWT
Malve and Uzsoy [33]		✓	✓		✓									Lmax
Cheng et al. [5]		✓	✓									✓		TWT
Chiang et al. [7]		✓	✓		✓									Lmax
Kurz and Mason [28]	✓		✓		✓							✓		TWT
Li and Qiao [29]	✓		✓									✓		TWT
Li et al. [31]		✓	✓									✓		TWT
Chiang et al. [6]		✓	✓									✓		TWT
Guo et al. [18]		✓	✓						✓					TWT
Kim et al. [25]		✓	✓											TT
Li et al. [30]		✓	✓		✓									TWT/Cmax
Jia and Jiang [22]		✓	✓		✓							✓		TWT/Lmax/TWCT
Bilyk et al. [2]		✓	✓									✓		TWT
Vimala Rani and Mathirajan [54]	✓			✓	✓							✓		TWT
Vimala Rani and Mathirajan [56]	✓			✓	✓									TWT
Vimala Rani and Mathirajan [55]	✓			✓	✓							✓		TWT/NT/OTD rate/TE/Lmax
This Research Study	✓			✓	✓	✓	✓	✓	✓	✓	✓	✓		TWT/OTD rate/TE/Lmax

B Batches

Index:

j 1...N for jobs
b 1...K for batches
f 1...G for families

Parameters:

A First time availability of the DF
BC Batch capacity of DF
 RT_j Release time of a job 'j'
 DD_j Due-date of a job 'j'
 W_j Weight of a job 'j'
 PT_f Processing time of a family 'f'
 FA_{jf} Family association for a job, equals 1 if job 'j' belong to family 'f'; 0 otherwise

Decision variables:

JB_{jb} 1 if job 'j' is processed in a batch 'b'; 0 Otherwise
 FB_{fb} 1 if family 'f' is processed in a batch 'b'; 0 Otherwise

Dependent variables:

CB_b Completion time of a batch 'b'
 RB_b Release time of a batch 'b'
 PB_b Processing time of a batch 'b'
 CJ_j Completion time of job 'j'
 T_j Tardiness of job 'j'
 L_j Lateness of job 'j'
 L_{max} Maximum Lateness among 'N' jobs
 EL_j Absolute deviations of completion times from the due-date of job 'j'
 UP_j Unit penalty of job 'j'
OTDrate On-time delivery rate among 'N' jobs

4.1 Proposed Model I: formulation of mathematical model for DS-DF to minimize TWT

$$\text{Minimize } \sum_{j=1}^N W_j * T_j \quad (1)$$

Subject to:

$$\sum_{f=1}^G FB_{fb} \leq 1 \quad \forall b \in [1, K] \quad (2)$$

$$\sum_{f=1}^G FB_{f,b-1} \geq \sum_{f=1}^G FB_{fb} \quad \forall b \in [2, K] \quad (3)$$

$$(JB_{jb} * FA_{jf}) \leq FB_{fb} \quad \forall j \in [1, N]; \forall b \in [1, K]; \forall f \in [1, G] \quad (4)$$

$$\sum_{j=1}^N (JB_{jb} * FA_{jf}) \geq FB_{fb} \quad \forall f \in [1, G]; \forall b \in [1, K] \quad (5)$$

$$\sum_{b=1}^K JB_{jb} = 1 \quad \forall j \in [1, N] \quad (6)$$

$$\sum_{j=1}^N JB_{jb} \leq BC \quad \forall b \in [1, K] \quad (7)$$

$$RTB_b \geq A \quad b = 1 \quad (8)$$

$$RB_b \geq (RT_j * JB_{jb}) \quad \forall j \in [1, N]; \quad \forall b \in [1, K] \quad (9)$$

$$RB_b \geq CB_{b-1} \quad \forall b \in [2, K] \quad (10)$$

$$PB_b = \sum_{f=1}^G PT_f * FB_{fb} \quad \forall b \in [1, K] \quad (11)$$

$$CB_b = RB_b + PB_b \quad \forall b \in [1, K] \quad (12)$$

$$CJ_j \geq CB_b - (\text{BigM}) * (1 - JB_{jb}) \quad \forall b \in [1, K]; \quad \forall j \in [1, N] \quad (13)$$

$$T_j \geq CJ_j - DD_j \quad \forall j \in [1, N] \quad (14)$$

$$JB_{jb} \in \{0, 1\} \quad \forall j \in [1, N]; \quad \forall b \in [1, K] \quad (15)$$

$$FB_{fb} \in \{0, 1\} \quad \forall f \in [1, G]; \quad \forall b \in [1, K] \quad (16)$$

The objective function presented in (1) is to minimize the TWT. Constraints (2) to (7) are related to batch formation of diffusion furnace. Constraint (2) ensures that incompatibility nature of diffusion furnace. Constraint (3) is used to form a batch sequentially. Constraint (4) ensures that one-to-one mapping between ‘job is assigned to a batch’ and corresponding ‘family is assigned to a batch’. Constraint (5) assures that a batch must contain at least one job of a family that is assigned to that batch. Constraint (6) forces that a job can be processed in only one batch. Constraint (7) is the DF capacity constraint.

Constraints (8) to (10) are related to release time of a batch. Constraint (8) guarantees that the release time of first batch is not less than the first time availability of the DF. In addition to that, release time of any batch should be greater than the maximum arrival time of jobs in that batch and the completion time of previous batch. For that Constraint (9) and Constraint (10) are formed. Constraint (11) is used to calculate the dependent variable processing time of batch. It can be calculated by processing time of family that constitutes in that batch. Constraint (12) calculates the dependent variable completion time of batch. It is the sum of release time and processing time of batch. Constraint (13) is used to calculate the completion time of job. Constraint (14) is used to calculate the tardiness of jobs (that is, positive lateness). It can be calculated as the difference between completion time of job and its due-date. Finally, the constraint (15) and Constraint (16) assign the binary values to the decision variables.

As most of the constraints in Model I remain the same for the other three Proposed Models, only the modified as well as additional constraints required for each of the other three Proposed Models are presented here.

4.2 Proposed Model II: formulation of mathematical model for DS-DF to maximize OTD rate

The objective of this proposed model is maximizing the on-time delivery (OTD) rate. For this, out of ‘N’ jobs how many jobs are completed on or before its due-date needs to be identified. Accordingly, the modified and additional constraints on the Proposed Model I are developed and the same are given here.

Modified constraint

$$\text{Maximize } OTD/N \quad (17)$$

Additional constraints

$$UP_j \geq \min(1, T_j) \quad \forall j \in [1, N] \quad (18)$$

$$OTD = \left[N - \left(\sum_{j=1}^N UP_j \right) \right] \quad (19)$$

Objective function Constraint (1) in the Proposed Model I needs to be modified as Constraint (17). The Constraint (17) is used to maximize the on-time delivery (OTD) rate. It is defined as the ratio between number of jobs completed on or before its due-date (that is, total on-time delivery jobs) and the total number of jobs (N). Total on-time delivery jobs can be calculated by subtracting number of tardy (late) jobs from the total number of jobs (N). For that, Constraint (18) is used to assign the unit penalty to each tardy job and Constraint (19) is used to compute total on-time delivery jobs.

4.3 Proposed Model III: formulation of mathematical model for DS-DF to minimize TE/L

In the proposed model III, the objective is to minimize the absolute difference between completion time of job and its due-date for each job. Hence, the TWT calculation (Constraint 1) and tardiness computation (Constraint 14) for each job in Proposed Model I need to be modified as Constraint (20) and Constraint (21) respectively. Accordingly, the modified constraints are given below.

Modified constraint

$$\text{Minimize } \sum_{j=1}^N EL_j \quad (20)$$

$$EL_j \geq |CT_j - DD_j| \quad \forall j \in [1, N] \quad (21)$$

Constraint (20) is to minimize the total earliness/lateness. It can be defined as sum of absolute difference between completion time of job and its due-date. Constraint (21) is used to compute the dependent variable earliness/lateness (EL) by calculating the absolute difference between completion time of job and its due-date.

4.4 Proposed Model IV: formulation of mathematical model for DS-DF to minimize Lmax

In order to minimize maximum lateness, first task that needs to be calculated is the lateness value for each job and then compute maximum lateness among 'N' jobs, considered for scheduling. Similar to previous model, Constraint (1) and Constraint (14) in the Proposed Model I needs to be modified as Constraint (22) and Constraint (23) respectively. Accordingly, the modified and additional constraints required for the Proposed Model IV are as follows:

Modified constraint

$$\text{Minimize } L_{\max} \quad (22)$$

$$L_j \geq CT_j - DD_j \quad \forall j \in [1, N] \quad (23)$$

Additional constraint

$$L_{\max} = \max_{j=1 \text{ to } N}(L_j) \quad (24)$$

Constraint (22) is to minimize $L_{\max}$ (that is, maximum lateness) among ‘N’ jobs. Lateness of each job ‘j’ is defined as the difference between completion time of job and its due-date. Constraint (23) is used to calculate the dependent variable ‘lateness of job’. Finally, constraint (24) is used to calculate maximum lateness among ‘N’ jobs.

4.5 Validation of the proposed mathematical models for DS-DF

For validating the proposed four mathematical models for DS-DF, a numerical problem is developed with job characteristics: job code, family identification, release time, due-date, weight and the same is given in Table 3. The diffusion furnace capacity is measured in terms of number jobs it holds. In general, diffusion furnace (DF) can accommodate between 6 and 12 standard lots (jobs) and for numerical problem it is assumed to be 2 jobs. The initial availability of DF in the planning period for processing a batch is assumed to be second hour. For generating each of the proposed mathematical models for any given data, suitable LINGO set code are developed. Due to the brevity of the paper, the LINGO set code developed for Proposed Model I is only presented in Appendix 1.

For the numerical problem presented in Table 3 each of the proposed models are generated using the corresponding developed LINGO set code and solved the proposed models using LINGO solver. The optimal solution obtained from LINGO solver for each of the scheduling objective: TWT, OTD rate, TE/L, and Lmax are presented in Table 4. From the analysis of the optimal solution given in Table 4 the validation of the proposed models are verified.

Table 3 A numerical problem to validate the proposed models for DS-DF

Job code	Family identification	Release time	Due-date	Weight
J1	5	4	29	4
J2	1	3	18	3
J3	4	2	34	7
J4	3	3	20	4
J5	2	4	21	5
J6	2	2	18	9
J7	3	2	35	3
J8	5	3	34	7
J9	4	6	27	8
J10	4	4	21	3

Table 4 Scheduling objective wise the optimal solution for DS-DF for the numerical problem

Objective: Minimizing TWT													
Batch	Job Code	Optimal Batch			Job								
		RTB	PTB	CTB	DD	W	T	E	L	WT=W*T	UPJ	EL	
B1	J2	3	2	5	18	3	0			0			
B2	J5	5	4	9	21	5	0			0			
	J6				18	9	0			0			
B3	J3	9	16	25	34	7	0			0			
	J9				27	8	0			0			
B4	J4	25	10	35	20	4	15			60			
	J7				35	3	0			0			
B5	J1	35	20	55	29	4	26			104			
	J8				34	7	21			147			
B6	J10	55	16	71	21	3	50			150			
TWT for the Numerical Example										461			
Objective: Maximizing OTD rate													
B1	J2	3	2	5	18	3					0		
B2	J5	5	4	9	21	5					0		
	J6				18	9					0		
B3	J3	9	16	25	34	7					0		
	J9				27	8					0		
B4	J7	25	10	35	35	3					0		
B5	J8	35	20	55	34	7					1		
B6	J10	55	16	71	21	3					1		
B7	J4	71	10	81	20	4					1		
B8	J1	81	20	101	29	4					1		
OTD rate for the Numerical Example (10- $\sum$ unit penalty)/10										0.6			
Objective: Minimizing TE/L													
B1	J2	4	2	6	18	3		12	0				12
B2	J9	6	16	22	27	8		5	0				5
	J10				21	3		0	1				1
B3	J5	22	4	26	21	5		0	5				5
	J6				18	9		0	8				8
B4	J4	26	10	36	20	4		0	16				16
	J7				35	3		0	1				1
B5	J1	36	20	56	29	4		0	27				27
	J8				34	7		0	22				22
B6	J3	56	16	72	34	7		0	38				38
TE/L for the Numerical Example													135
Objective: Minimizing Lmax													
B1	J4	3	10	13	20	4			0				
	J7				35	3			0				
B2	J1	13	20	33	29	4			4				
	J8				34	7			0				
B3	J9	33	16	49	27	8			22				
	J10				21	3			28				
B4	J2	49	2	51	18	3			33				
B5	J5	51	4	55	21	5			34				
	J6				18	9			37				
B6	J3	55	16	71	34	7			37				
Lmax for the Numerical Example										37			

4.6 Computational complexity of the proposed mathematical models for DS-DF

To understand the computational complexity of the proposed models fifteen problem instances including the one in Table 3 are generated using Turbo C by considering release time of jobs, due-date of jobs, and weight of jobs are in the uniform interval of [1–8], [1–40], and [1–10] respectively. Particularly five problem instances are generated by increasing the number of jobs from six jobs to ten jobs and the next ten problem instances are generated by keeping the number of jobs as 10. In addition to that, we consider each problem has five incompatible job-families with the processing time of 2 h, 4 h, 10 h, 16 h, and 20 h respectively. Further, it is assumed that batch capacity is 2 jobs and the initial available time of diffusion furnace in the planning period for processing a batch is the second hour.

All the fifteen problem instances are solved using LINGO solver to get the optimal solution for the scheduling objectives: TWT, OTD rate, TE/L, and Lmax by generating the appropriate proposed mathematical model using corresponding LINGO set code, developed in this study. Accordingly, the optimal value and computational time required to get the optimal solution for each of the scheduling objectives: TWT, OTD rate, TE/L, and Lmax, for each of these fifteen problem instances are observed and presented in Table 5.

As the computational time for small-scale problem is large (Table 5) and does not follow any pattern when the problem size increases as well as when the problem size kept as constant, the proposed mathematical model is not amenable to solve large-scale problem. Thus, the alternate method is required to get the efficient solution in a meager time. It is observed from the literature that simple greedy heuristic algorithms (GHA), mostly based on dispatching rule(s), are used to obtain efficient solution as quick as possible in scheduling [48]. Hence, this study focuses GHA based on dispatching rules particularly, ‘due-date based dispatching rules (DDR)’ for DS-DF and the same is discussed in the next section.

5 Proposed Greedy Heuristic Algorithm (GHA) based on DDR (GHA-DDR) for DS-DF

Scheduling single diffusion furnace (a single batch-processing machine), involves two interrelated decisions: (1) batch formation, and (2) batch selection at each decision time “t”. First decision (or) batch formation is decided by Job-Priority-Index, and the second decision (or) batch selection is decided by Batch-Priority-Index.

This study proposes GHA based on DDR for dynamic scheduling of a single diffusion furnace in the presence of dynamic job arrivals with incompatible-job families and non-agreeable release time and due-date to optimize the due-date based scheduling objectives: TWT, OTD rate, TE/L, and Lmax. Accordingly, the systematic procedure of the proposed GHA-DDR is as follows:

Table 5 Scheduling objective wise optimal value and computational time for DS-DF

Problem instances	No. of jobs	Scheduling objective						
		TWT		OTD Rate		TE/L		Lmax
		Optimal TWT	Computational time (hh:mm:ss)	Optimal OTD rate	Computational time (hh:mm:ss)	Optimal TE/L	Computational time (hh:mm:ss)	
1	6	111	0:00:04	0.667	0:00:20	40	0:00:05	21
2	7	111	0:00:28	0.714	0:01:15	60	0:00:50	21
3	8	222	0:01:30	0.625	0:14:31	71	0:04:54	21
4	9	311	0:05:22	0.667	2:21:41	91	0:22:52	26
5	10	461	0:16:18	0.6	12:17:13	135	1:59:13	37
6	10	112	0:18:20	0.7	15:13:32	102	4:18:41	20
7	10	247	0:16:07	0.7	16:58:24	73	1:10:51	24
8	10	325	0:21:45	0.7	4:59:51	80	0:46:31	25
9	10	200	0:07:17	0.9	0:30:59	54	0:51:09	14
10	10	182	0:05:21	0.8	4:48:25	59	0:51:06	16
11	10	120	0:14:04	0.8	7:09:16	68	2:16:46	13
12	10	429	0:28:13	0.6	15:30:50	116	1:41:53	30
13	10	76	0:12:48	0.8	7:11:52	84	3:41:35	8
14	10	56	0:18:05	0.7	13:02:21	52	0:46:01	15
15	10	120	0:07:30	0.8	8:24:24	37	0:19:19	9
								0:07:57

- Step 1: Captures the next available time of diffusion furnace (say T , $T > 0$). Also captures all the job characteristics from work-in-process for each available job (that is, jobs that are arrived on or before ' T '). Then classify the available jobs based on the job-characteristic: Family and named as Job-Family
- Step 2: For each Job-Family, calculate the **Job-Priority-Index** using DDR for *each available jobs*
- Step 3: *Sort* the available jobs in each Job-Family based on the **Job-Priority-Index**
- Step 4: Form a *temporary batch* from each Job-Family by selecting a set of jobs from the top of each Job-Family by respecting batch capacity constraint (that is, the number of jobs in each temporary batch should not exceed DF capacity)
- Step 5: Compute *Batch-Priority-Index* using DDR for each temporary batch formed from each Job-Family
- Step 6: Compare *Batch-Priority-Index* of each temporary batch and select the temporary batch, which has *maximum Batch-Priority-Index*
- Step 7: *Allocate* the selected batch, decided from step 6 to DF. Capture the processing time of the family in the allocated batch and store it as the processing time of allocated batch
- Step 8: *Compute* TWT, OTD rate, TE/L , and L_{max} of allocated batch (es)
- Step 9: *Update the work-in-process of the jobs* by excluding the jobs allocated to the DF. Also, update the next available time (T) of DF as $T = T + \text{Processing time of allocated batch in Step 7}$
- Step 10: *Repeat* Step 1 until all jobs are scheduled

5.1 Twenty proposed variants of GHA-DDR for DS-DF

By using various due-date based dispatching rules (DDR) for computing Job-Priority-Index and Batch-Priority-Index, in this paper, twenty variants of GHA-DDR have been proposed. The due-date based dispatching rules considered for computing Job-Priority-Index and Batch-Priority-Index are presented here.

5.1.1 DDR considered for Computing Job-Priority Index

Earliest Due-Date (EDD): This rule is the most popular simple due-date based dispatching rule. It is a function of job characteristic 'due-date'. This rule is under static category since it is not time dependent. Using EDD the Job-Priority-Index of job ' j ' can be calculated as follows.

$$\text{Job-Priority-Index (j)} = DD_j$$

Flow Due-Date (FDD): Jayamohan and Rajendran [21] propose this rule. It is a static rule. FDD of job ' j ' for operation ' m ' is a sum of release time of job ' j ' (RT_j) and the total processing time of job ' j ' till the operation ' m '. Accordingly, the Job-Priority-Index of job ' j ' for operation ' m ' is as follows.

$$\text{Job-Priority-Index (j)} = RT_j + \sum_{q=1}^m PT_{jq}$$

where PT_{jq} = Process time required for job 'j' for the operation 'q', m = Current operation.

As this study concerns about only one operation (i.e.) diffusion operation, the total processing time of job 'j' till the operation 'm' (here $m = 1$) is equal to the processing time of job 'j' (PT_j) for diffusion operation. Accordingly, this study modified the Job-Priority-Index of job 'j' using FDD is as follows.

$$\text{Job-Priority-Index (j)} = RT_j + PT_j$$

Operational due-date (ODD): This rule is adopted from Jayamohan and Rajendran [21]. This rule is also a static due-date based dispatching rule. It is a function of job characteristics: "release time" and "due-date". The Job-Priority-Index of job 'j' using ODD is as follows.

$$\text{Job-Priority-Index (j)} = RT_j + c * \sum_{q=1}^m PT_{jq}$$

where c = due-date allowance factor (and it is assumed to be equal to 3).

As we mentioned, this study concerns about only one operation [(i.e.) diffusion operation], the 'm' (current operation) value is set to one. Now the Job-Priority-Index becomes as follows.

$$\text{Job-Priority-Index (j)} = RT_j + c * PT_j$$

Modified Operational Due-Date (MOD): Like ODD, this rule is also adopted from Jayamohan and Rajendran [21]. It is a dynamic rule since it is time dependent. In general, using MOD, the Job-Priority-Index of job 'j' for operation 'm' can be calculated as follows.

$$\text{Job-Priority-Index (j)} = \max \left\{ \left(RT_j + c * \sum_{q=1}^m PT_{jq} \right), \tau + PT_{jq} \right\}$$

where τ = time at which the dispatching decision is made.

This study modified the above equation in order to calculate Job-Priority-Index of job 'j' for only one operation (diffusion operation) by setting $m = 1$. Accordingly, the Job-Priority-Index of job 'j' for DF using MOD is as follows.

$$\text{Job-Priority-Index (j)} = \max \{ (RT_j + c * PT_j), \tau + PT_j \}$$

Critical ratio (CR): It is a composite dispatching rule. It is defined as the ratio of earliest due-date and shortest remaining processing time [39]. It is also a dynamic rule since it depends on time. The critical ratio (CR) of a job is calculated as the difference between the job due-date and the today's date divided by its total remaining processing time. We considered time at which the diffusion furnace became free as

today's date and processing time of job 'j' as total shop time remaining since this study focus only single operation. The formula for calculating the Job-Priority-Index using CR is as follows.

$$\text{Job-Priority-Index (j)} = (\text{DD}_j - \text{Today's date}) / \text{Total shop time remaining}$$

A ratio less than 1.0 implies that the job is behind schedule, and a ratio greater than 1.0 implies that the job is ahead of schedule.

Minimum slack (MS) first: This is a simple and dynamic dispatching rule. It is a measure of urgency for a given job. The formula for calculating Job-Priority-Index using MS first rule is as follows.

$$\text{Job-Priority-Index (j)} = (\text{DD}_j - t - \text{PT}_j)^+$$

where t = time at which the dispatching decision is made (i.e.) Time at which the diffusion furnace became free. In general, the notation $\mathbf{x}^+$ represents $\max(\mathbf{x}, \mathbf{0})$.

This rule implies that at some point in time job 'j' may have a higher priority than job 'k' and at some later point in time jobs 'j' and 'k' may have the same priority.

Cost OVER Time (COVERT): This rule is a composite dispatching rule. It is a combination of two simple dispatching rules: weighted shortest processing time (WSPT) and minimum slack (MS) First. When a job is projected to be tardy (i.e., its slack is zero) its priority index reduces to WSPT. On the other hand, if a job is expected to be very early where the slack exceeds an estimation of delay cost, the priority index value for job increases linearly with decreases in slack [15]. Since time dependent attribute ('t') is in this rule, it is under the dynamic category. The formula for calculating Job-Priority-Index is as follows.

$$\text{Job-Priority-Index (j)} = \left(\frac{W_j}{\text{PT}_j} \right) * \left[1 - \left((\text{DD}_j - t - \text{PT}_j)^+ / \sum_{j=1}^N \text{PT}_j \right) \right]^+$$

Apparent tardiness cost (ATC): It is a composite and dynamic dispatching rule. This rule combines the weighted shortest processing time (WSPT) rule and the minimum slack (MS) first rule. In this rule, an ATC ranking index is computed for each remaining job when a machine (here, Diffusion Furnace) is free. It is a function of few/all of the parameters namely, the time "t", at which the machine becomes free, processing time (PT_j), due-date (DD_j), the weight (W_j) and release time (RT_j) of the remaining jobs. Morton and Rachamadugu [41] proposed three apparent priority rules for scheduling a serial batching problem to minimize weighted tardiness. That study empirically commented that one of the apparent priority rules performed better than other two versions of the apparent priority rules. Then Vepsalainen and Morton [53] called this better performing apparent priority rule as ATC rule. Subsequently many researchers either introduced newer versions of ATC rules or extended the existing ATC rules for various problem configurations of serial and parallel batching problems. This study considered ten different ATC rules proposed/used/extended by various researchers in their research studies (Table 6). All these

ten ATC rules differ only in the way the ATC ranking index, that is, Job-Priority-Index is computed.

5.1.2 DDR considered for Computing Batch-Priority Index

In this study, Batch-Priority-Index is computed based on either Weighted Tardiness of Batch (WTB) rule or Batch Apparent Tardiness Cost (BATC) rule. The computational details of these rules are as follows:

WTB rule: According to this rule, the batch-priority-index is computed as follows:

$$\text{Batch-Priority-Index}(b) = \sum_{j=1}^K \{ (CT_b - DD_j) * W_j \}$$

where K =number of jobs in a temporary batch ‘b’, CT_b =max (DF available time, longest ready time of all jobs in the batch ‘b’)+processing time of family in that batch ‘b’, DD_j =due-date of job ‘j’, W_j =weight of job ‘j’.

Batch apparent tardiness cost (BATC): It is a dynamic dispatching rule. BATC ranking index is computed for each batch when a batch processing machine (BPM) is free. It is a function of few/all of the parameters namely, the time “t” at which the machine becomes free, processing time (PT_j), due-date (DD_j), weight (W_j), release time (RT_j) of the jobs in that batch and the release time (RTB_b) of the batch. Like ATC, many researchers proposed/used/extended various BATC rules for computing Batch-Priority-Index towards selecting a batch for batch processing machine. This study considered five different BATC rules proposed/used/extended by various researchers (Table 7).

Based on the DDR considered in this study for computing Job-Priority-Index and Batch-Priority-Index, twenty variants of GHA-DDR have been proposed. Each of the proposed twenty variants of GHA-DDR for DS-DF differs only in computing Job-Priority-Index (Step 2) for batch formation, sorting order (Step 3) that is, ascending order, or descending order of Job-Priority-Index, and computing Batch-Priority-Index (Step 5) for batch selection. The modified steps (that is, Step 2, Step 3 and Step 5) for each of these twenty proposed variants of GHA-DDR are presented in Table 8. All the twenty variants of GHA-DDR are implemented in Turbo C. The performance evaluations of the twenty proposed variants of GHA-DDR for each of the scheduling objectives: TWT, OTD rate, TE/L, and Lmax are carried out. The details on these are presented in the next section.

Table 6 Ten different existing ATC rules considered for computing *Job-Priority-Index* in the proposed variants of GHA-DDR

Name of the ATC Rule	Job-Priority-Index(j) =	Source
ATC-Morton	$\left(\frac{W_j}{PT_j}\right) * \exp^{-\left[\frac{k * (\max(0, DD_j - PT_j - t))}{\left(\frac{1}{N} * \sum_{i=1}^N PT_i\right)}\right]}$	Morton and Rachamadugu [41]
ATC-Vepsalainen	$\left(\frac{W_j}{PT_j}\right) * \exp^{-\left[\frac{\max(0, DD_j - PT_j - t)}{k * \left(\frac{1}{N} * \sum_{i=1}^N PT_i\right)}\right]}$	Vepsalainen and Morton [53]
ATC-Farhad 1	$\left(\frac{1}{PT_j}\right) * \exp^{-\left[\frac{\max(0, DD_j - (PT_j + t))}{\left(\frac{1}{N} * \sum_{i=1}^N PT_i\right)}\right]}$	Farhad and Laya [13]
ATC-Farhad 2	$\left(\frac{1}{PT_j + DD_j}\right) * \exp^{-\left[\frac{\max(0, DD_j - (PT_j + t))}{\left(\frac{1}{N} * \sum_{i=1}^N PT_i\right)}\right]}$	
ATC-Farhad 3	$\left(\frac{1}{PT_j}\right) * \exp^{-\left[\frac{\max(0, DD_j - (PT_j + \max(PT_j, t)))}{\left(\frac{1}{N} * \sum_{i=1}^N PT_i\right)}\right]}$	
ATC-Farhad 4	$\left(\frac{1}{PT_j + DD_j}\right) * \exp^{-\left[\frac{\max(0, DD_j - (PT_j + \max(PT_j, t)))}{\left(\frac{1}{N} * \sum_{i=1}^N PT_i\right)}\right]}$	
ATC-Bala	$\left(\frac{W_j}{PT_j}\right) * \exp^{-\left[\frac{\max(0, DD_j - PT_j - t)}{k * \left(\frac{1}{N} * \sum_{i=1}^N PT_i\right)}\right]}$	Balasubramanian <i>et al.</i> [1]
ATC-Monch	$\left(\frac{W_j}{PT_j}\right) * \exp^{-\left[\frac{\max(0, DD_j - PT_j + (RT_j - t))}{k * \left(\frac{1}{N} * \sum_{i=1}^N PT_i\right)}\right]}$	Monch <i>et al.</i> [40]
ATC-Li	$\left(\frac{W_j}{PT_j}\right) * \left(\exp^{-\left[\frac{\max(0, DD_j - PT_j - t + (RT_j - t)^+)}{k * \left(\frac{1}{N} * \sum_{i=1}^N PT_i\right)}\right]}\right)$	Li <i>et al.</i> [30]
ATC-Vimala	$\left(\frac{W_j}{PT_j}\right) * f(W_j, T_j) * \exp^{-\left[\frac{(DD_j - PT_j - t + (RT_j - t)^+)}{k * \left(\frac{1}{N} * \sum_{i=1}^N PT_i\right)}\right]}$ Where $f(W_j, T_j) = 1$ if $(RTB_b + PT_j - DD_j) \leq 0$; $W_j / (RTB_b + PT_j - DD_j)$ otherwise	Vimala Rani and Mathirajan [56]

Note: The value of the parameter 'K' in each of the ATC Rule is assumed to be 0.5 [Li *et al.* (2010)]

6 Performance evaluation of twenty proposed variants of GHA-DDR for DS-DF and analysis

A series of computational experiments is carried out to understand the efficiency of the twenty proposed variants of GHA-DDR for DS-DF. For having appropriate computational experiments, experimental design, benchmark solution procedure, and performance measure are required. The details of these are presented in this section.

Experimental design: Based on observation made in literature, there are six important problem parameters: number of jobs (N), arrival time of jobs (RT_j), processing time of jobs (PT_j), due-date of jobs (DD_j), family (f) and priority of jobs (W_j) could affect the performance of the variants of GHA-DDR. In the proposed experimental design, it is assumed that value of parameters except number of jobs, family and processing time is drawn from uniform distributions. The uniform distribution was chosen because it is a relatively high variance distribution, which would allow GHA-DDR to be tested under conditions relatively unfavourable to them. Further to control the tightness of due-dates, the following relationship is used during the generation of problem instances for obtaining a value for each job's due-date:

$$\text{due_date} \geq \text{release_time} + \text{processing_time}$$

Furthermore, batch capacity is assumed to be 6. The processing time of job-family 1, 2, 3, 4, and 5 is assumed to be 2, 4, 10, 16, and 20 h respectively. In addition, it

Table 7 Five different existing BATC rules considered for computing *Batch-Priority-Index* in the proposed variants of GHA-DDR

Name of the BATC Rule	Batch-Priority-Index(b)=	Source
BATC-Mehta	$\left(\frac{1}{p_{T_i}}\right) * \exp^{-\left[\sum_{j \in b} \max(0, DD_j - p_{T_j} - t) / k * \left(\frac{1}{N} * \sum_{l=1}^N p_{T_l}\right)\right]}$	Mehta and Uzsoy [37]
BATC-Bala	$\sum_{j \in b} \left\{ \left(\frac{w_j}{p_{T_j}}\right) * \exp^{-\left[\max(0, DD_j - p_{T_j} - t) / k * \left(\frac{1}{N} * \sum_{l=1}^N p_{T_l}\right)\right]} \right\}$	Balasubramanian <i>et al.</i> [1]
BATC-Monch	$\sum_{j \in b} \left\{ \left(\frac{w_j}{p_{T_j}}\right) * \left(\exp^{-\left[\max(0, DD_j - p_{T_j} - t + (RTB_b - t)^+\right) / k * \left(\frac{1}{N} * \sum_{l=1}^N p_{T_l}\right)\right]} * (nbj/B) \right\}$	Monch <i>et al.</i> [40]
BATC-Li	$\sum_{j \in b} \left\{ \left(\frac{w_j}{p_{T_j}}\right) * \left(\exp^{-\left[\max(0, DD_j - p_{T_j} - t + (RT_j - t)^+\right) / k * \left(\frac{1}{N} * \sum_{l=1}^N p_{T_l}\right)\right]} * \min((nbj/B), 1) \right\}$	Li <i>et al.</i> [30]
BATC-Vimala	$\sum_{j \in b} \left\{ \left(\frac{w_j}{p_{T_j}}\right) * f(W_j, T_i) * \exp^{-\left[\left(DD_j - p_{T_j} - t + (RT_j - t)^+\right) / \left(k * \left(\frac{1}{N} * \sum_{l=1}^N p_{T_l}\right)\right)\right]} * (nbj/B) \right\}$ Where $f(W_j, T_i) = 1$ if $(RT_B_b + p_{T_j} - DD_j) \leq 0$; $W_j / (RTB_b + p_{T_j} - DD_j)$ otherwise	Vimala Rani and Mathirajan [56]

is assumed that 10%, 30%, 40%, 10%, and 20% of the time job arrives for diffusion operation correspond to family 1, 2, 3, 4, and 5 respectively with the corresponding probability of 0.1, 0.3, 0.4, 0.1, and 0.2. Based on these, a summary of experimental design is developed and given in Table 9.

Benchmark solution procedure: For small-scale problem, optimal value obtained from the proposed mathematical model, presented in Table 5, is considered as a benchmark solution (BS) for performance evaluation of the twenty proposed variants of GHA-DDR. Whereas for large-scale problem, estimated optimal value, which is calculated based on the benchmark solution procedure discussed in Rardin and Uzsoy [46] is considered as benchmark solution.

Performance measures: Since the efficiency of the twenty proposed variants of GHA-DDR may vary over a range of problem instances, the efficiency of twenty proposed variants of GHA-DDR is analyzed both empirically and statistically.

For *empirical analysis*, this study uses the standard performance measure: Average Relative Percentage Deviation (ARPD). For *statistical analysis*, this study conducts Friedman test and Nemenyi multiple comparison test. The details of these measures are given below.

Average relative percentage deviation (ARPD): For each problem instance and for each variants of GHA-DDR, we compute the relative percentage deviation (RPD) w.r.t. benchmark solution (BS) using Eq. (25).

$$RPD_{ij} = [D_{ij}/BS_i] * 100 \quad (25)$$

where i : problem instances and $i \in [1, 270]$, j : variant of GHA-DDR and $j \in [1, 20]$, BS_i : equal to optimal value of 'ith' problem instance, for small-scale problem, BS_i : equal to estimated optimal value of 'ith' problem instance, for large-scale problem, D_{ij} : feasible solution (GHA-DDR_{ij})- BS_i , for TWT, TE/L, and Lmax (minimization problem), D_{ij} : BS_i —feasible solution (GHA-DDR_{ij}), for OTD rate (maximization problem), RPD_{ij} : relative percentage deviation of 'jth' variant of GHA-DDR for 'ith' problem instances.

Next, compute the average of RPD (ARPD) for each variants of GHA-DDR over the number of problem instances planned in each problem configurations as well as the total number of problem instances in the experimental design as per the Eq. (26).

$$ARPD_j = \sum_{i=1}^N RPD_{ij}/N \quad (26)$$

where $ARPD_j$: average relative percentage deviation of 'jth' variant of GHA-DDR, $N:=10$ when ARPD is computed problem configuration wise, and $N:=270$ when ARPD is computed considering entire problem instances.

For *statistical analysis*, this study conducts Friedman test as per Settouti et al. [49] to determine whether there are any significant differences among the twenty

Table 8 Modified steps in the twenty proposed variants of GHA-DDR for DS-DF

No.	Proposed variants of GHA-DDR	Step 2: Batch formation using Job-Priority-Index based on	Step 3: Sorting order of Job- Priority-Index	Step 5: Batch selection using Batch-Priority-Index based on
1	GHA-DDR1(EDD,WTB)	EDD	Ascending	Weighted Tardiness of Batch (WTB)
2	GHA-DDR2 (FDD,WTB)	FDD	Ascending	WTB
3	GHA-DDR3 (ODD,WTB)	ODD	Ascending	WTB
4	GHA-DDR4 (MOD,WTB)	MOD	Ascending	WTB
5	GHA-DDR5 (CR,WTB)	CR	Ascending	WTB
6	GHA-DDR6 (MS,WTB)	MS First	Ascending	WTB
7	GHA-DDR7 (COVERT,WTB))	COVERT	Descending	WTB
8	GHA-DDR8 (ATC-Morton, WTB)	ATC-Morton	Descending	WTB
9	GHA-DDR9 (ATC-Vepsalainen, WTB)	ATC-Vepsalainen	Descending	WTB
10	GHA-DDR10 (ATC-Farhad 1, WTB)	ATC-Farhad 1	Descending	WTB
11	GHA-DDR11 (ATC-Farhad 2, WTB)	ATC-Farhad 2	Descending	WTB
12	GHA-DDR12 (ATC-Farhad 3, WTB)	ATC-Farhad 3	Descending	WTB
13	GHA-DDR13 (ATC-Farhad 4, WTB)	ATC-Farhad 4	Descending	WTB
14	GHA-DDR14 (EDD, BATC-Mehta)	EDD	Ascending	BATC-Mehta
15	GHA-DDR15 (ATC-Bala, BATC-Bala)	ATC-Bala	Descending	BATC-Bala
16	GHA-DDR16 (ATC-Monch, BATC-Monch)	ATC-Monch	Descending	BATC-Monch
17	GHA-DDR17 (ATC-Li, BATC-Li)	ATC-Li	Descending	BATC-Li
18	GHA-DDR18 (ATC-Li, BATC-Monch)	ATC-Li	Descending	BATC-Monch
19	GHA-DDR19 (ATC-Vimala, BATC-Monch)	ATC-Vimala	Descending	BATC-Monch
20	GHA-DDR20 (ATC-Vimala, BATC-Vimala)	ATC-Vimala	Descending	BATC-Vimala

proposed variants of GHA-DDR. *Friedman test* is the non-parametric test to determine whether there are significant differences among the algorithms considered over given sets of data. It determines the ranks of the algorithms for each individual data set. In general, it assigns highest rank (Rank 1) to minimum value. This test can only detect whether there is any significant differences among the algorithm (here, GHA-DDR). It is not in a position to point out which pair of algorithms differs significantly. For that, this study conducts post hoc test particularly, *Nemenyi* multiple comparison test.

6.1 Performance of the twenty proposed variants of GHA-DDR: empirical analysis

The efficiency of each of the twenty proposed variants of GHA-DDR for DS-DF with respect to each of the scheduling objectives considered in this study is analysed empirically in comparison with (1) optimal value obtained from the corresponding

Table 9 A summary of the proposed experimental design

Parameters	Levels	Number of levels
Number of jobs (N)	25,50,100	3
Release time of jobs (RT_j)	[1, 8], [1, 16], [1, 24]	3
Due-date of jobs (DD_j)	[1, 40], [1,60], [1,80]	3
Family (f)	[1, 5]	1
Weight (W_j) [Low, High]	[1, 10]	1
Family processing time (PT_f) and the probability of job being the family 'f'	2 with a probability of 0.1 4 with a probability of 0.3 10 with a probability of 0.4 16 with a probability of 0.1 20 with a probability of 0.1	1
Number of problem configurations ($3*3*3*1*1*1$)		27
Number of instances per configuration		10
Total Problem Instances		270

mathematical models for small-scale problem, and (2) estimated optimal value for large-scale problem, using the standard performance measure: ARPD.

6.1.1 Empirical analysis in comparison with optimal value

As the mathematical models need more time to get the optimal solution, the performance analysis of twenty proposed variants of GHA-DDR is carried out in comparison with optimal value obtained (Table 5), in this study, for the fifteen small-scale problem instances for each of the scheduling objectives: TWT, OTD rate, TE/L, and Lmax. Each of the fifteen problem instances are given as input to each of the twenty proposed variants of GHA-DDR and obtained the value of the due-date based scheduling objectives, considered in this study.

For each problem instances and for each of the due-date based scheduling objectives considered in the study, relative percentage deviation (RPD) of the solution obtained from each of the twenty proposed variants of GHA-DDR w.r.t. corresponding optimal value, is calculated using Eq. (25) to understand how the performance of each of the proposed variants of GHA-DDR is close to the optimal solution. The computed RPD score for each of the twenty proposed variants of GHA-DDR with respect to each of the scheduling objectives: TWT, OTD rate, TE/L, and Lmax is presented in Appendices 2 to 5 respectively.

Using the RPD score given in Appendices 2 to 5, the average of RPD (ARPD) over fifteen problem instances for each of the twenty proposed variants of GHA-DDR with respect to each of the scheduling objectives: TWT, OTD rate, TE/L, and Lmax is computed and the same is represented in Fig. 1. From the Fig. 1, it is observed that, (1) GHA-DDR15 is providing the solution close to the optimal:

TWT and OTD rate, (2) GHA-DDR14 is providing the solution close to the optimal: TE/L, and (3) ten variants: GHA-DDR1 to GHA-DDR6 and GHA-DDR10 to GHA-DDR13 are providing the solution close to the optimal: Lmax. We cannot stoutly wind up and generalize this inference for all problem sizes, as these inferences are obtained on small-scale problems. Hence, this study carried out the performance evaluation of the twenty proposed variants of GHA-DDR in comparison with estimated optimal value, for large-scale problem.

6.1.2 Empirical analysis in comparison with estimated optimal value

A computer code in Turbo C is developed and generated 270 test problems, considering the proposed experimental design presented in Table 9. Each of these 270 problem instances are solved by each of the twenty proposed variants of GHA-DDR and obtained the corresponding solution for each of the scheduling objectives: TWT, OTD rate, TE/L, and Lmax. So, for each instances, we have twenty feasible solutions for each of the scheduling objectives considered in this study. Using the twenty feasible solutions with respect to each of the scheduling objectives for each of the 270 problem instances an estimated optimal value is obtained following the procedure discussed in Rardin and Uzsoy [46].

Now we compute RPD score for each of the twenty variants of GHA-DDR, considering the solution obtained from the variants of GHA-DDR and the estimated optimal solution, using Eq. (25) for each of the 270 problem instances. Using the RPD score obtained, we compute ARPD score for each of the 27 problem configurations and for each of the twenty proposed variants of GHA-DDR using the Eq. (26) over ten problem instances.

Due to brevity of the paper, ARPD score for each of the 27 problem configurations for each of the twenty proposed variants of GHA-DDR w.r.t. the scheduling objective TWT is only presented in Appendix 6 and for the remaining objectives: OTD rate, TE/L, and Lmax, the ARPD score computed are not reported in this study. Further, irrespective of the problem configurations, the ARPD score for each of the twenty proposed variants of GHA-DDR, over 270 problem instances, with respect to each of the scheduling objectives: TWT, OTD rate, TE/L, and Lmax is computed and is presented in Table 10 and in Fig. 2.

From Table 10 and Fig. 2, it is observed that GHA-DDR15 to GHA-DDR20 are relatively performing better than the other variants of GHA-DDR for most of the scheduling objectives considered in this study. The possible reason for the outperforming variant: GHA-DDR15 to GHA-DDR20 may be due to the fact that, all these variants use composite dispatching rule, particularly ATC based composite dispatching rule, for both decisions: constructing the batch and selecting the batch. Particularly, it is observed that (a) GHA-DDR18 and GHA-DDR19 are better variants for the scheduling objective: TWT and OTD rate, and (b) GHA-DDR16 and GHA-DDR18 are better variants for the scheduling objective: TE/L and Lmax.

Here, GHA-DDR16, GHA-DDR18 and GHA-DDR19 computes Job-Priority-Index based on ATC rule proposed by Monch et al. [40], Li et al. [30] and Vimala

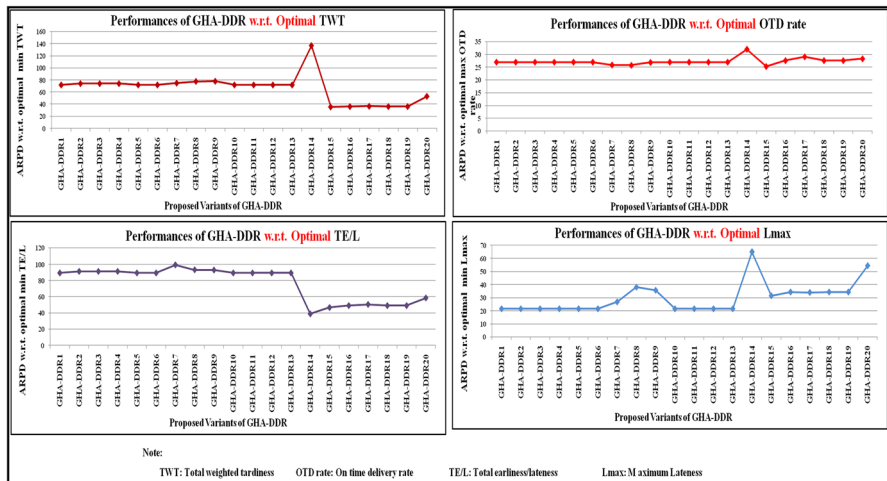


Fig. 1 Performance the proposed variants of GHA-DDR in comparison with optimal: TWT, OTD rate, TE/L, and Lmax, over 15 small-scale problem instances

Rani and Mathirajan [56] respectively and Batch-Priority-Index based on BATC rule proposed by Monch et al. [40]. The ATC rule given in Monch et al. [40], Li et al. [30], and Vimala Rani and Mathirajan [56] considers the ‘release time of job’ while computing Job-Priority-Index. Hence, these ATC rules could yield an accurate value of the remaining time or slack. Further, BATC rule given in Monch et al. [40] considers the ‘release time of batch’ while selecting the batch, which ensures the batch that has the minimum starting time gets first priority to schedule. Therefore, it reduces the lateness of jobs in the current batch as well as the upcoming batch. Due to these reasons, these variants become the better performing variants.

6.2 Performance of the twenty proposed variants of GHA-DDR: statistical analysis

This study conducts Friedman test to determine whether there are any significant differences among the twenty variants of GHA-DDR while optimizing the due-date based scheduling objectives, considered in this study. Accordingly, the statistically measurable hypotheses defined are as follows:

H1₀ There is no significant difference among the twenty variants of GHA-DDR while minimizing TWT.

H2₀ There is no significant difference among the twenty variants of GHA-DDR while maximizing OTD rate.

H3₀ There is no significant difference among the twenty variants of GHA-DDR while minimizing TE/L.

Table 10 ARPD score of each of the twenty proposed variants of GHA-DDR for the Scheduling objective: TWT, OTD rate, TE/L, and Lmax

Proposed variants of GHA-DDR	ARPD score of each of the proposed variants of GHA-DDR for the scheduling objective			
	TWT	OTD rate	TE/L	Lmax
GHA-DDR1	117.93	52.71	30.68	21.83
GHA-DDR2	136.94	38.24	47.49	33.36
GHA-DDR3	136.94	38.24	47.49	33.36
GHA-DDR4	136.94	38.24	47.49	33.36
GHA-DDR5	117.93	52.71	30.68	21.83
GHA-DDR6	118.39	50.27	31.47	25.45
GHA-DDR7	95.09	38.93	48.47	35.39
GHA-DDR8	132.22	61.18	65.83	44.54
GHA-DDR9	142.44	64.67	62.07	37.17
GHA-DDR10	118.38	50.27	31.47	25.45
GHA-DDR11	117.93	52.71	30.68	21.83
GHA-DDR12	118.47	50.27	31.57	25.49
GHA-DDR13	117.93	52.71	30.68	21.83
GHA-DDR14	197.81	66.82	25.01	36.53
GHA-DDR15	11.09	14.15	6.98	23.49
GHA-DDR16	7.37	11.52	4.70	17.90
GHA-DDR17	10.98	15.05	6.09	21.18
GHA-DDR18	3.84	9.04	5.36	18.88
GHA-DDR19	3.73	6.43	6.86	22.58
GHA-DDR20	18.92	18.31	14.56	31.64

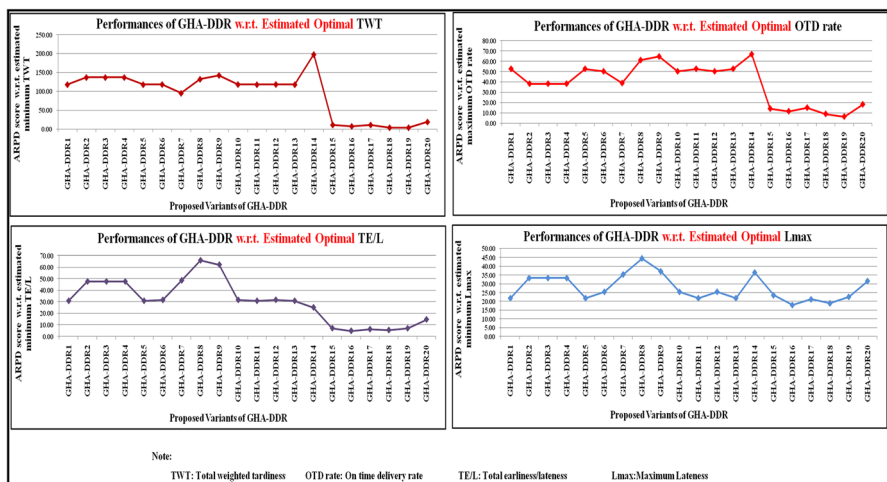


Fig. 2 Performance the proposed variants of GHA-DDR in comparison with estimated optimal: TWT, OTD rate, TE/L, and Lmax, over 270 large-scale problem instances

H₄₀ There is no significant difference among the twenty variants of GHA-DDR while minimizing Lmax.

The alternative hypothesis for each of the above measurable hypothesis (that is, null hypothesis) is the negation of the corresponding null hypothesis. The four hypotheses defined here are tested using Friedman test. The Friedman rank for the twenty variants of GHA-DDR w.r.t each of the scheduling objectives considered in this study and the Friedman test results are given in Tables 11 and 12 respectively.

From Table 11, it is observed that

- GHA-DDR18 and GHA-DDR19 are better performing variants as they have top rank for the scheduling objective: TWT and OTD rate. [here, the top rank is 1 for minimization objective: TWT, and 20 for maximization objective: OTD rate]
- GHA-DDR16 to GHA-DDR18 are better performing variants as they have top rank for the scheduling objective: TE/L and Lmax

This endorses the inferences obtained from the empirical analysis discussed previously. Further, from Table 12, it is observed that P value for all the hypotheses is less than 0.05. Therefore, there is a strong evidence to reject all the four null hypotheses (H10 and H40). Hence, there is a significant difference among twenty variants of GHA-DDR while optimizing each of the due-date based scheduling objectives: TWT, OTD rate, TE/L, and Lmax. However, Friedman test does not indicate which variant(s) really drives the significant difference among the twenty variants of GHA-DDR. To know this, we conduct Nemenyi post hoc test, which does pair-wise comparison of the twenty variants on the performance w.r.t. each objective. Accordingly, the defined 190 measurable hypotheses for Nemenyi post hoc test for each of the scheduling objectives are as follows:

H₀ There is no significant difference between ‘ith’ variant of GHA-DDR and ‘jth’ variant of GHA-DDR while minimizing TWT/OTD rate/TE/L/Lmax.

Where $i = 2$ to 20, $j = 1$ to $i - 1$.

The alternative hypothesis for each of the measurable hypothesis (that is, null hypothesis) is the negation of the corresponding null hypothesis.

The P values obtained from Nemenyi post hoc test for the scheduling objective TWT, OTD rate, TE/L, and Lmax are given in Tables 13, 14, 15 and 16 respectively. From these tables, it is observed that

- For the scheduling objective: TWT and OTD rate, the better performing variants: GHA-DDR15 to GHA-DDR20 are significantly differ from GHA-DDR1 to GHA-DDR14 since these P values are less than 0.05. The Nemenyi post hoc test clearly indicates that there is no significant difference between the top rank better performing variants: GHA-DDR18 and GHA-DDR19. Further, there is no significant difference between GHA-DDR16 and GHA-DDR18.

Table 11 Friedman rank for the twenty variants of GHA-DDR w.r.t. scheduling objective: TWT, OTD rate, TE/L, and Lmax

Proposed Variants of GHA-DDR	Ranking for the scheduling objective			
	TWT	OTD rate	TE/L	Lmax
GHA-DDR1	13.05	6.84	10.01	7.78
GHA-DDR2	15.21	11.88	15.26	12.41
GHA-DDR3	15.21	11.88	15.26	12.41
GHA-DDR4	15.21	11.88	15.26	12.41
GHA-DDR5	13.05	6.84	10.01	7.78
GHA-DDR6	12.79	8.23	9.99	9.89
GHA-DDR7	9.55	11.78	15.73	13.14
GHA-DDR8	12.97	5.42	17.77	16.12
GHA-DDR9	14.19	3.98	16.62	14.85
GHA-DDR10	12.78	8.23	9.98	9.89
GHA-DDR11	13.05	6.84	10.01	7.78
GHA-DDR12	12.81	8.23	10.04	9.94
GHA-DDR13	13.05	6.84	10.01	7.78
GHA-DDR14	13.96	4.72	8.75	10.60
GHA-DDR15	3.30	15.66	5.57	10.25
GHA-DDR16	4.37	16.02	4.89	7.91
GHA-DDR17	4.16	15.31	5.29	9.11
GHA-DDR18	2.86	16.69	5.42	8.15
GHA-DDR19	3.12	17.48	6.06	9.57
GHA-DDR20	5.29	15.26	8.07	12.19

Table 12 Friedman test results for the scheduling objective: TWT, OTD rate, TE/L, and Lmax

Test statistics	Friedman test results for the scheduling objective			
	TWT	OTD rate	TE/L	Lmax
N	270	270	270	270
Chi square	3280.004	3178.294	2618.006	1053.643
Degrees of freedom	19	19	19	19
Asymp. Sig. (<i>P</i> value)	0.000	0.000	0.000	0.000

- For the scheduling objective: TE/L, GHA-DDR15 to GHA-DDR19 are significantly differ from GHA-DDR1 to GHA-DDR14. Further, the Nemenyi post hoc test clearly indicates that there is no significant difference between the top rank better performing variants: GHA-DDR16 to GHA-DDR18.
- Finally, for the scheduling objective Lmax, the better performing variants: GHA-DDR1, GHA-DDR5, GHA-DDR11, GHA-DDR13 and GHA-DDR16 to GHA-DDR18 are significantly differing from other variants and there is no significant difference between these better performing variants.

From the above discussions, it appears that the variants: GHA-DDR16 to GHA-DDR18 are better performing variants for all the due-date based scheduling objectives, considered in this study.

Overall, from the empirical and statistical analysis, it is observed that, the variants (that is, GHA-DDR15 to GHA-DDR20), which uses ATC and BATC dispatching rule for batch formation and batch selection respectively are better performing variants for most of the scheduling objectives considered in this study. Particularly, it is observed that GHA-DDR16, which uses ATC and BATC rule proposed by Monch et al. [40], GHA-DDR17, which uses ATC and BATC rule proposed by Li et al. [30], and GHA-DDR18, which uses ATC rule proposed by Li et al. [30], and BATC rule proposed by Monch et al. [40] are better variants for all the scheduling objectives considered in this study. These ATC and BATC rules consider ‘release time of job’ and ‘release time of batch’ respectively while computing job-priority index and batch-priority index respectively.

7 Conclusion

From the literature it is observed that dispatching rules are widely used in semiconductor manufacturing industry because they (a) are perform reasonably well in a wide range of environments, (b) are relatively easy to implement and understand, (c) need only minimal computational time, and (d) can easily manage dynamic changes. Further, the literature review revealed that though there has been extensive scheduling research on comparing the dispatching rules in job-shop, flow-shop, open-shop, flexible manufacturing system, and single machine with unit capacity environments, there is no such analysis of dispatching rules on dynamic scheduling of diffusion furnace with incompatible Job-families, and non-agreeable release times and due-dates with the objective of optimizing various due-date based scheduling objectives: TWT, OTD rate, TE/L, and Lmax.

This study first, proposes mathematical models for dynamic scheduling of a single diffusion furnace to optimize each of the scheduling objectives: TWT, OTD rate, TE/L, and Lmax individually. The proposed mathematical models are validated by solving small-scale problem. Further, fifteen small-scale problem instances are solved by each of the proposed mathematical models for understanding the computational intractability. Due to the computational difficulty in obtaining solution for large-scale problem in reasonable time, this study proposes twenty variants of simple greedy heuristic algorithm (GHA) based on due-date based dispatching rules (DDR) such as EDD, FDD, ODD, MOD, CR, MS first, COVERT, & set of ATC and/or BATC.

The performance analyses of twenty proposed variants of GHA-DDR for DS-DF with respect to each of the scheduling objectives considered in this study was analysed empirically in comparison with (a) optimal value obtained from the proposed mathematical models for 15 problem instances (small-scale problem), and (b) estimated optimal value for 270 problem instances (large-scale problem), using the standard performance measure: ARPD, and statistically by Friedman test and Nemenyi (Post-hoc) multiple comparison test.

From the empirical and statistical analyses, it is observed that the variants, which use ATC and BATC dispatching rules for batch formation and batch selection

Table 13 Pair wise comparison of twenty variants of GHA-DDR for the scheduling objective: TWT

Proposed variants of GHA-DDR	P values for pair wise comparison between variants of GHA-DDR									
	GHA-DDR1	GHA-DDR2	GHA-DDR3	GHA-DDR4	GHA-DDR5	GHA-DDR6	GHA-DDR7	GHA-DDR8	GHA-DDR9	GHA-DDR10
GHA-DDR2	0.00363									
GHA-DDR3	0.00363	1.00000								
GHA-DDR4	0.00363	1.00000	1.00000							
GHA-DDR5	1.00000	0.00363	0.00363	0.00363						
GHA-DDR6	1.00000	0.00034	0.00034	0.00034	1.00000					
GHA-DDR7	1.1e-09	7.1e-14	7.1e-14	7.1e-14	1.1e-09	3.6e-08				
GHA-DDR8	1.00000	0.00185	0.00185	0.00185	1.00000	1.00000	3.2e-09			
GHA-DDR9	0.77765	0.90160	0.90160	0.90160	0.77765	0.38610	2.5e-13	0.66676	0.37826	
GHA-DDR10	1.00000	0.00032	0.00032	0.00032	1.00000	1.00000	3.9e-08	1.00000	0.77765	1.00000
GHA-DDR11	1.00000	0.00363	0.00363	0.00363	1.00000	1.00000	1.1e-09	1.00000	0.42083	1.00000
GHA-DDR12	1.00000	0.00043	0.00043	0.00043	1.00000	1.00000	2.7e-08	1.00000	0.77765	1.00000
GHA-DDR13	1.00000	0.00363	0.00363	0.00363	1.00000	1.00000	1.1e-09	1.00000	1.00000	0.73332
GHA-DDR14	0.96847	0.60782	0.60782	0.60782	0.96847	0.74094	1.7e-13	0.92952	1.00000	<2e-16
GHA-DDR15	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16
GHA-DDR16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	1.9e-13	<2e-16	<2e-16	<2e-16
GHA-DDR17	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	1.8e-13	<2e-16	<2e-16	<2e-16
GHA-DDR18	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16
GHA-DDR19	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16
GHA-DDR20	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	<2e-16	2.2e-13	<2e-16	<2e-16	<2e-16

Table 13 (continued)

Proposed variants of GHA- DDR	P values for pair wise comparison between variants of GHA-DDR									
	GHA-DDR11	GHA-DDR12	GHA-DDR13	GHA-DDR14	GHA-DDR15	GHA-DDR16	GHA-DDR17	GHA-DDR18	GHA-DDR19	
GHA-DDR2										
GHA-DDR3										
GHA-DDR4										
GHA-DDR5										
GHA-DDR6										
GHA-DDR7										
GHA-DDR8										
GHA-DDR9										
GHA-DDR10										
GHA-DDR11										
GHA-DDR12	1.00000									
GHA-DDR13	1.00000	1.00000								
GHA-DDR14	0.96847	0.77290	0.96847							
GHA-DDR15	<2e-16	<2e-16	<2e-16	<2e-16						
GHA-DDR16	<2e-16	<2e-16	<2e-16	<2e-16	0.85254					
GHA-DDR17	<2e-16	<2e-16	<2e-16	<2e-16	0.98141	1.00000				
GHA-DDR18	<2e-16	<2e-16	<2e-16	<2e-16	1.00000	0.24919	0.55049			
GHA-DDR19	<2e-16	<2e-16	<2e-16	<2e-16	1.00000	0.61066	0.88607	1.00000		
GHA-DDR20	<2e-16	<2e-16	<2e-16	<2e-16	0.01371	0.96289	0.78469	0.00033	0.00326	

Table 14 Pair wise comparison of twenty variants of GHA-DDR for the scheduling objective: OTD Rate

Proposed variants of GHA-DDR	<i>P</i> values for pair wise comparison between variants of GHA-DDR									
	GHA-DDR1	GHA-DDR2	GHA-DDR3	GHA-DDR4	GHA-DDR5	GHA-DDR6	GHA-DDR7	GHA-DDR8	GHA-DDR9	GHA-DDR10
GHA-DDR2	1.8e-13									
GHA-DDR3	1.8e-13	1.0000								
GHA-DDR4	1.8e-13	1.0000	1.0000							
GHA-DDR5	1.0000	1.8e-13	1.8e-13	1.8e-13						
GHA-DDR6	0.4100	1.5e-10	1.5e-10	1.5e-10	0.4100					
GHA-DDR7	2.5e-13	1.0000	1.0000	1.0000	2.5e-13	6.2e-10				
GHA-DDR8	0.3577	<2e-16	<2e-16	<2e-16	0.3577	6.3e-06	<2e-16			
GHA-DDR9	3.6e-06	<2e-16	<2e-16	<2e-16	3.6e-06	<2e-16	0.3402	6.3e-06	2.2e-13	
GHA-DDR10	0.4100	1.5e-10	1.5e-10	1.5e-10	0.4100	1.0000	6.2e-10	0.3577	3.6e-06	0.4100
GHA-DDR11	1.0000	1.8e-13	1.8e-13	1.8e-13	1.0000	0.4100	2.5e-13	6.3e-06	2.2e-13	1.0000
GHA-DDR12	0.4100	1.5e-10	1.5e-10	1.5e-10	0.4100	1.0000	6.2e-10	0.3577	3.6e-06	0.4100
GHA-DDR13	1.0000	1.8e-13	1.8e-13	1.8e-13	1.0000	0.4100	2.5e-13	0.3577	0.9968	1.0e-09
GHA-DDR14	0.0049	<2e-16	<2e-16	<2e-16	0.0049	1.0e-09	<2e-16	0.9984	<2e-16	<2e-16
GHA-DDR15	<2e-16	2.2e-11	2.2e-11	2.2e-11	<2e-16	<2e-16	4.8e-12	<2e-16	<2e-16	<2e-16
GHA-DDR16	<2e-16	3.2e-13	3.2e-13	3.2e-13	<2e-16	<2e-16	2.2e-13	<2e-16	<2e-16	<2e-16
GHA-DDR17	<2e-16	3.2e-09	3.2e-09	3.2e-09	<2e-16	<2e-16	7.9e-10	<2e-16	<2e-16	<2e-16
GHA-DDR18	<2e-16	1.7e-13	1.7e-13	1.7e-13	<2e-16	<2e-16	1.4e-13	<2e-16	<2e-16	<2e-16
GHA-DDR19	<2e-16	1.5e-13	1.5e-13	1.5e-13	<2e-16	<2e-16	4.9e-14	<2e-16	<2e-16	<2e-16
GHA-DDR20	<2e-16	6.2e-09	6.2e-09	6.2e-09	<2e-16	<2e-16	1.6e-09	<2e-16	<2e-16	<2e-16

Table 14 (continued)

Proposed variants of GHA- DDR	P values for pair wise comparison between variants of GHA-DDR									
	GHA-DDR11	GHA-DDR12	GHA-DDR13	GHA-DDR14	GHA-DDR15	GHA-DDR16	GHA-DDR17	GHA-DDR18	GHA-DDR19	
GHA-DDR2										
GHA-DDR3										
GHA-DDR4										
GHA-DDR5										
GHA-DDR6										
GHA-DDR7										
GHA-DDR8										
GHA-DDR9										
GHA-DDR10										
GHA-DDR11										
GHA-DDR12	0.4100									
GHA-DDR13	1.0000	0.4100								
GHA-DDR14	0.0049	1.0e-09	0.0049							
GHA-DDR15	<2e-16	<2e-16	<2e-16	<2e-16						
GHA-DDR16	<2e-16	<2e-16	<2e-16	<2e-16	1.0000					
GHA-DDR17	<2e-16	<2e-16	<2e-16	<2e-16	1.0000	0.9977				
GHA-DDR18	<2e-16	<2e-16	<2e-16	<2e-16	0.8893	0.9991	0.4100			
GHA-DDR19	<2e-16	<2e-16	<2e-16	<2e-16	0.0457	0.3207	0.0033	0.9934		
GHA-DDR20	<2e-16	<2e-16	<2e-16	<2e-16	1.0000	0.9947	1.0000	0.3402	0.0022	

Table 15 Pair wise comparison of twenty variants of GHA-DDR for the scheduling objective: TE/L

Proposed variants of GHA-DDR	<i>P</i> values for pair wise comparison between variants of GHA-DDR									
	GHA-DDR1	GHA-DDR2	GHA-DDR3	GHA-DDR4	GHA-DDR5	GHA-DDR6	GHA-DDR7	GHA-DDR8	GHA-DDR9	GHA-DDR10
GHA-DDR2	2.1e-13									
GHA-DDR3	2.1e-13	1.00000								
GHA-DDR4	2.1e-13	1.00000	1.00000							
GHA-DDR5	1.00000	2.1e-13	2.1e-13	2.1e-13						
GHA-DDR6	1.00000	1.9e-13	1.9e-13	1.9e-13	1.00000					
GHA-DDR7	4.0e-14	1.00000	1.00000	1.00000	4.0e-14	3.1e-14				
GHA-DDR8	<2e-16	0.00015	0.00015	0.00015	<2e-16	<2e-16	0.00966			
GHA-DDR9	<2e-16	0.44275	0.44275	0.44275	<2e-16	<2e-16	0.97161	0.77051		
GHA-DDR10	1.00000	1.8e-13	1.8e-13	1.8e-13	1.00000	1.00000	2.7e-14	<2e-16	<2e-16	
GHA-DDR11	1.00000	2.1e-13	2.1e-13	2.1e-13	1.00000	1.00000	4.0e-14	<2e-16	<2e-16	1.00000
GHA-DDR12	1.00000	1.6e-13	1.6e-13	1.6e-13	1.00000	1.00000	5.3e-14	<2e-16	<2e-16	1.00000
GHA-DDR13	1.00000	2.1e-13	2.1e-13	2.1e-13	1.00000	1.00000	4.0e-14	<2e-16	<2e-16	1.00000
GHA-DDR14	0.59926	<2e-16	<2e-16	<2e-16	0.59926	0.63894	<2e-16	<2e-16	<2e-16	0.64734
GHA-DDR15	1.6e-13	<2e-16	<2e-16	<2e-16	1.6e-13	1.7e-13	<2e-16	<2e-16	<2e-16	1.7e-13
GHA-DDR16	2.2e-13	<2e-16	<2e-16	<2e-16	2.2e-13	1.6e-13	<2e-16	<2e-16	<2e-16	1.6e-13
GHA-DDR17	2.1e-13	<2e-16	<2e-16	<2e-16	2.1e-13	2.2e-13	<2e-16	<2e-16	<2e-16	2.2e-13
GHA-DDR18	1.3e-13	<2e-16	<2e-16	<2e-16	1.3e-13	1.3e-13	<2e-16	<2e-16	<2e-16	1.4e-13
GHA-DDR19	1.7e-12	<2e-16	<2e-16	<2e-16	1.7e-12	2.5e-12	<2e-16	<2e-16	<2e-16	2.7e-12
GHA-DDR20	0.01951	<2e-16	<2e-16	<2e-16	0.01951	0.02347	<2e-16	<2e-16	<2e-16	0.02441

Table 15 (continued)

Proposed variants of GHA-DDR	P values for pair wise comparison between variants of GHA-DDR									
	GHA-DDR11	GHA-DDR12	GHA-DDR13	GHA-DDR14	GHA-DDR15	GHA-DDR16	GHA-DDR17	GHA-DDR18	GHA-DDR19	
GHA-DDR2										
GHA-DDR3										
GHA-DDR4										
GHA-DDR5										
GHA-DDR6										
GHA-DDR7										
GHA-DDR8										
GHA-DDR9										
GHA-DDR10										
GHA-DDR11										
GHA-DDR12	1.00000									
GHA-DDR13	1.00000	1.00000								
GHA-DDR14	0.59926	0.56486	0.59926							
GHA-DDR15	1.6e-13	1.6e-13	1.6e-13	8.0e-08						
GHA-DDR16	2.2e-13	2.1e-13	2.2e-13	6.7e-12	0.99891					
GHA-DDR17	2.1e-13	2.0e-13	2.1e-13	2.0e-09	1.00000	1.00000				
GHA-DDR18	1.3e-13	2.7e-13	1.3e-13	1.2e-08	1.00000	0.99997	1.00000			
GHA-DDR19	1.7e-12	1.2e-12	1.7e-12	2.4e-05	0.99999	0.73587	0.99440	0.99953		
GHA-DDR20	0.01951	0.01661	0.01951	0.99891	0.00016	8.0e-08	8.7e-06	3.7e-05	0.01194	

Table 16 Pair wise comparison of twenty variants of GHA-DDR for the scheduling objective: Lmax

Proposed variants of GHA-DDR	P values for pair wise comparison between variants of GHA-DDR									
	GHA-DDR1	GHA-DDR2	GHA-DDR3	GHA-DDR4	GHA-DDR5	GHA-DDR6	GHA-DDR7	GHA-DDR8	GHA-DDR9	GHA-DDR10
GHA-DDR2	2.6e-13									
GHA-DDR3	2.6e-13	1.00000								
GHA-DDR4	2.6e-13	1.00000	1.00000							
GHA-DDR5	1.00000	2.6e-13	2.6e-13	2.6e-13						
GHA-DDR6	0.00547	0.00013	0.00013	0.00013	0.00547					
GHA-DDR7	1.9e-13	0.99741	0.99741	0.99741	1.9e-13	3.4e-08				
GHA-DDR8	<2e-16	6.1e-11	6.1e-11	6.1e-11	<2e-16	<2e-16	8.8e-07			
GHA-DDR9	<2e-16	0.00031	0.00031	0.00031	<2e-16	2.3e-13	0.09050	0.57922		
GHA-DDR10	0.00547	0.00013	0.00013	0.00013	0.00547	1.00000	3.4e-08	<2e-16	2.3e-13	
GHA-DDR11	1.00000	2.6e-13	2.6e-13	2.6e-13	1.00000	0.00547	1.9e-13	<2e-16	<2e-16	0.00547
GHA-DDR12	0.00358	0.00022	0.00022	0.00022	0.00358	1.00000	6.5e-08	<2e-16	1.5e-13	1.00000
GHA-DDR13	1.00000	2.6e-13	2.6e-13	2.6e-13	1.00000	0.00547	1.9e-13	<2e-16	<2e-16	0.00547
GHA-DDR14	6.2e-06	0.04565	0.04565	0.04565	6.2e-06	0.99827	0.00011	2.0e-13	2.2e-13	0.99827
GHA-DDR15	0.00023	0.00342	0.00342	0.00342	0.00023	1.00000	2.5e-06	<2e-16	1.3e-13	1.00000
GHA-DDR16	1.00000	1.5e-13	1.5e-13	1.5e-13	1.00000	0.01510	2.2e-13	<2e-16	<2e-16	0.01510
GHA-DDR17	0.50178	1.6e-08	1.6e-08	1.6e-08	0.50178	0.99335	7.4e-13	<2e-16	3.1e-14	0.99335
GHA-DDR18	1.00000	2.1e-13	2.1e-13	2.1e-13	1.00000	0.07347	2.1e-13	<2e-16	<2e-16	0.07347
GHA-DDR19	0.05519	4.4e-06	4.4e-06	4.4e-06	0.05519	1.00000	4.6e-10	<2e-16	1.8e-13	1.00000
GHA-DDR20	1.7e-13	1.00000	1.00000	1.00000	1.7e-13	0.00111	0.94669	2.2e-12	3.3e-05	0.00111

Table 16 (continued)

Proposed variants of GHA- DDR	<i>P</i> values for pair wise comparison between variants of GHA-DDR									
	GHA-DDR11	GHA-DDR12	GHA-DDR13	GHA-DDR14	GHA-DDR15	GHA-DDR16	GHA-DDR17	GHA-DDR18	GHA-DDR19	
GHA-DDR2										
GHA-DDR3										
GHA-DDR4										
GHA-DDR5										
GHA-DDR6										
GHA-DDR7										
GHA-DDR8										
GHA-DDR9										
GHA-DDR10										
GHA-DDR11										
GHA-DDR12	0.00358									
GHA-DDR13	1.00000	0.00358								
GHA-DDR14	6.2e-06	0.99938	6.2e-06							
GHA-DDR15	0.00023	1.00000	0.00023	1.00000						
GHA-DDR16	1.00000	0.01022	1.00000	2.6e-05	0.00080					
GHA-DDR17	0.50178	0.98607	0.50178	0.27695	0.77765	0.70199				
GHA-DDR18	1.00000	0.05328	1.00000	0.00028	0.00598	1.00000	0.94271			
GHA-DDR19	0.05519	1.00000	0.05519	0.89861	0.99899	0.12006	1.00000	0.36025		
GHA-DDR20	1.7e-13	0.00176	1.7e-13	0.16899	0.01978	2.1e-13	2.8e-07	6.9e-13	5.1e-05	

respectively, yield consistently better solution, in comparison with other variants. Probably, these ATC rules could yield an accurate value of the remaining time or slack, while forming a batch and this BATC rule might ensures the batch that has the minimum starting time gets first priority to schedule. Moreover, within the better performing variants, the variants: GHA-DDR16 to GHA-DDR18, which use ATC rule having ‘release time of job’ for batch formation (Monch et al. [40] and Li et al. [30])) and BATC rule having ‘release time of batch’ for batch selection (Monch et al. [40]) yield better solution for all the scheduling objectives considered in this study. That is, the ATC and BATC rules considered in the better performing variants reduces the lateness of jobs in the current batch as well as the upcoming batch and due to these reasons, these variants: GHA-DDR16 to GHA-DDR18 become the better performing variants.

This study concerns about due-date based scheduling objectives but semiconductor manufacturing industry may be interested to deal with completion time based scheduling objective. Hence, the observation obtained in this study needs to be verified for completion time based scheduling objectives. Further, the findings obtained in this study considering single DF need to be verified for the case of multiple diffusion furnaces. These could be our immediate future research directions.

Appendix 1: A LINGO set code for the proposed mathematical model for DS-DF to minimize TWT

! Minimize Total Weighted Tardiness in DS-DF;

SETS:

```
JOBS=R_TIME,DUE_DATE,PENALTY,TARDINESS,CT_JOB;
FAMILIES=PR_TIME;
BATCHES=CT_BATCH,RT_BATCH,PT_BATCH;
D_SET_JF(JOBS,FAMILIES):JF;
D_SET_JB(JOBS,BATCHES):JB;
D_SET_FB(FAMILIES,BATCHES):FB;
```

ENDSETS

DATA:

```
JOBS=J1,J2,J3,J4,J5,J6,J7,J8,J9,J10;
FAMILIES=F1,F2,F3,F4,F5;
BATCHES=B1,B2,B3,B4,B5,B6,B7,B8,B9,B10;
R_TIME=5,8,8,4,8,1,2,6,7,4;
DUE_DATE=39,15,36,56,18,36,55,50,36,48;
PENALTY=10,4,9,10,5,10,7,9,10,9;
PR_TIME=2,4,10,16,20;
CAPACITY=2;
AVAILABILITY=2;
JF=@FILE('fam_ass9.ldr');
```

ENDDATA

! Objective function;

```
MIN=@SUM(JOBS(j):PENALTY(j)*TARDINESS(j));
```

! Constraint-2: At most one family can be processed in each batch;

```
@FOR(BATCHES(b):
    @SUM(FAMILIES(f):FB(f,b))<=1);
```

! Constraint-3: A batch can be formed only if its previous batch has been formed;

```
@FOR(BATCHES(b)b #GT# 1:
    @SUM(FAMILIES(f):FB(f,b-1))>=@SUM(FAMILIES(f):FB(f,b));
```

! Constraint-4: A job can be processed in a batch only if the family to which it belongs to is processed in that batch;

```
@FOR(JOBS(j):
    @FOR(FAMILIES(f):
        @FOR(BATCHES(b):
            JB(j,b)*JF(j,f)<=FB(f,b))));
```

! Constraint-5: If some family is assigned to a batch then that batch must contain at least one job of that family;

```
@FOR(FAMILIES(f):
    @FOR(BATCHES(b):
        @SUM(JOBS(j):JB(j,b)*JF(j,f))>=FB(f,b));
```


! Constraint-6: Jobs can be processed in only one batch;

```
@FOR(JOBS(j):
    @FOR(D_SET_JB(j,b):@BIN(JB(j,b))));
@FOR(JOBS(j):
    @SUM(BATCHES(b):JB(j,b))=1);
```

! Constraint-7: Batch capacity;

```
@FOR(BATCHES(b):
    @SUM(D_SET_JB(j,b):JB(j,b))<=CAPACITY);
```

! Constraint-8: Release time of the first batch on a BPM is greater than the BPM available time;
RT_BATCH(1)>=AVAILABILITY;

! Constraint-9: Release time of a batch must not be less than the release time of any of jobs constituting the batch;

```
@FOR(BATCHES(b):
    @FOR(JOBS(j):
        RT_BATCH(b)>=R_TIME(j)*JB(j,b));
```

! Constraint-10: Release time of batch is greater than the completion time of previous batch;

```
@FOR(BATCHES(b)|b#GT#1:
    RT_BATCH(b)>=CT_BATCH(b-1));
```

! Constraint-11: Processing time of batch is the processing time of family in that batch;

```
@FOR(BATCHES(b):
    PT_BATCH(b)=@SUM(FAMILIES(f):PR_TIME(b)*FB(f,b));
```

! Constraint-12: Completion time of batch is the sum of ready time and processing time of that batch;

```
@FOR(BATCHES(b):
    CT_BATCH(b)=RT_BATCH(b)+PT_BATCH(b));
```

! Constraint-13: Completion time of job is the completion time of the batch where that job is assigned;

```
@FOR(JOBS(j):
    @FOR(BATCHES(b):
        CT_JOB(j)>=CT_BATCH(b)-10000*(1-JB(j,b))));
```

! Constraint-14: Tardiness of job is the difference between completion time of job and its due-date;

```
@FOR(JOBS(j):
    TARDINESS(j)=CT_JOB(j)-DUE_DATE(j));
```

! Constraint-15 & 16: Assign binary values to the decision variable;

```
@FOR(D_SET_JB:@BIN(JB));
@FOR(D_SET_FB:@BIN(FB));
```

END

Appendix 2: Performance analysis of each of the twenty proposed variants of GHA-DDR in comparison with optimal TWT based on Relative Percentage Deviation (RPD)

Problem instances	Optimal TWT	RPD of each of the proposed variants of GHA-DDR in comparison with optimal TWT									
		GHA-DDR1	GHA-DDR2	GHA-DDR3	GHA-DDR4	GHA-DDR5	GHA-DDR6	GHA-DDR7	GHA-DDR8	GHA-DDR9	GHA-DDR10
1	111	0.00	0.00	0.00	0.00	0.00	0.00	0.00	68.47	68.47	0.00
2	111	0.00	0.00	0.00	0.00	0.00	0.00	0.00	68.47	68.47	0.00
3	222	16.22	16.22	16.22	16.22	16.22	16.22	16.22	0.00	0.00	16.22
4	311	3.54	3.54	3.54	3.54	3.54	3.54	3.54	43.41	43.41	3.54
5	461	34.06	49.89	49.89	49.89	34.06	34.06	27.55	29.28	32.75	34.06
6	112	308.93	308.93	308.93	308.93	308.93	308.93	308.93	308.93	308.93	308.93
7	247	78.14	78.14	78.14	78.14	78.14	78.14	78.14	47.37	47.37	78.14
8	325	18.77	18.77	18.77	18.77	18.77	18.77	18.77	11.08	11.08	18.77
9	200	11.00	28.50	28.50	28.50	11.00	11.00	63.50	11.00	11.00	11.00
10	182	6.59	6.59	6.59	6.59	6.59	6.59	6.59	6.59	6.59	6.59
11	120	83.33	83.33	83.33	83.33	83.33	83.33	83.33	83.33	83.33	83.33
12	429	77.62	77.62	77.62	77.62	77.62	77.62	77.16	80.19	80.19	77.62
13	76	125.00	125.00	125.00	125.00	125.00	125.00	125.00	92.11	92.11	125.00
14	56	225.00	225.00	225.00	225.00	225.00	225.00	225.00	225.00	225.00	225.00
15	120	91.67	91.67	91.67	91.67	91.67	91.67	91.67	91.67	91.67	91.67
Problem instances	RPD of each of the proposed variants of GHA-DDR in comparison with optimal TWT										
	GHA-DDR11	GHA-DDR12	GHA-DDR13	GHA-DDR14	GHA-DDR15	GHA-DDR16	GHA-DDR17	GHA-DDR18	GHA-DDR19	GHA-DDR20	
1	0.00	0.00	0.00	8.11	9.91	9.91	9.91	9.91	9.91	109.01	
2	0.00	0.00	0.00	236.94	9.91	15.32	15.32	15.32	15.32	15.32	
3	16.22	16.22	16.22	5.86	8.11	10.81	10.81	10.81	10.81	27.03	
4	3.54	3.54	3.54	47.59	5.79	13.50	13.50	13.50	13.50	31.83	
5	34.06	34.06	34.06	34.06	36.23	36.23	36.23	36.23	36.23	48.59	
6	308.93	308.93	308.93	192.86	12.50	12.50	12.50	12.50	12.50	30.36	
7	78.14	78.14	78.14	47.37	23.08	23.08	23.08	23.08	23.08	23.08	
8	18.77	18.77	18.77	4.92	7.69	7.69	7.69	7.69	7.69	7.69	
9	11.00	11.00	11.00	319.00	8.00	1.50	8.00	1.50	1.50	44.50	
10	6.59	6.59	6.59	6.59	6.59	6.59	6.59	6.59	6.59	0.00	
11	83.33	83.33	83.33	335.00	73.33	73.33	75.00	73.33	73.33	127.50	
12	77.62	77.62	77.62	23.54	23.54	23.54	23.54	23.54	23.54	23.54	
13	125.00	125.00	125.00	496.05	15.79	15.79	15.79	15.79	15.79	15.79	
14	225.00	225.00	225.00	271.43	271.43	271.43	271.43	271.43	271.43	271.43	
15	91.67	91.67	91.67	23.33	23.33	23.33	23.33	23.33	23.33	23.33	

Appendix 3: Performance analysis of each of the twenty proposed variants of GHA-DDR in comparison with optimal OTD rate based on RPD

Problem instances	Optimal OTD rate	RPD of each of the proposed variants of GHA-DDR in comparison with optimal OTD rate									
		GHA-DDR1	GHA-DDR2	GHA-DDR3	GHA-DDR4	GHA-DDR5	GHA-DDR6	GHA-DDR7	GHA-DDR8	GHA-DDR9	GHA-DDR10
1	0.667	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	0.714	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
3	0.625	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
4	0.670	17.08	17.08	17.08	17.08	17.08	17.08	17.08	17.08	17.08	17.08
5	0.600	16.67	16.67	16.67	16.67	16.67	16.67	16.67	16.67	33.33	16.67
6	0.700	71.43	71.43	71.43	71.43	71.43	71.43	71.43	85.71	85.71	71.43
7	0.700	42.86	42.86	42.86	42.86	42.86	42.86	42.86	28.57	28.57	42.86
8	0.700	28.57	28.57	28.57	28.57	28.57	28.57	28.57	14.29	14.29	28.57
9	0.900	44.44	44.44	44.44	44.44	44.44	44.44	44.44	44.44	44.44	44.44
10	0.800	12.50	12.50	12.50	12.50	12.50	12.50	12.50	12.50	12.50	12.50
11	0.800	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00
12	0.600	66.67	66.67	66.67	66.67	66.67	66.67	50.00	50.00	50.00	66.67
13	0.800	25.00	25.00	25.00	25.00	25.00	25.00	25.00	37.50	37.50	25.00
14	0.700	28.57	28.57	28.57	28.57	28.57	28.57	28.57	28.57	28.57	28.57
15	0.800	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00

Problem instances	RPD of each of the proposed variants of GHA-DDR in comparison with optimal OTD rate									
	GHA-DDR11	GHA-DDR12	GHA-DDR13	GHA-DDR14	GHA-DDR15	GHA-DDR16	GHA-DDR17	GHA-DDR18	GHA-DDR19	GHA-DDR20
1	0.00	0.00	0.00	25.00	0.00	0.00	0.00	0.00	0.00	25.00
2	0.00	0.00	0.00	60.00	0.00	20.00	20.00	20.00	20.00	20.00
3	0.00	0.00	0.00	0.00	0.00	20.00	20.00	20.00	20.00	20.00
4	17.08	17.08	17.08	17.08	17.08	33.67	33.67	33.67	33.67	17.08
5	16.67	16.67	16.67	16.67	50.00	50.00	50.00	50.00	50.00	33.33
6	71.43	71.43	71.43	71.43	42.86	42.86	42.86	42.86	42.86	28.57
7	42.86	42.86	42.86	28.57	42.86	42.86	42.86	42.86	42.86	42.86
8	28.57	28.57	28.57	42.86	28.57	28.57	28.57	28.57	28.57	28.57
9	44.44	44.44	44.44	66.67	33.33	11.11	33.33	11.11	11.11	44.44
10	12.50	12.50	12.50	12.50	12.50	12.50	12.50	12.50	12.50	0.00
11	25.00	25.00	25.00	12.50	25.00	25.00	25.00	25.00	25.00	37.50
12	66.67	66.67	66.67	33.33	33.33	33.33	33.33	33.33	33.33	33.33
13	25.00	25.00	25.00	37.50	37.50	37.50	37.50	37.50	37.50	37.50
14	28.57	28.57	28.57	42.86	42.86	42.86	42.86	42.86	42.86	42.86
15	25.00	25.00	25.00	12.50	12.50	12.50	12.50	12.50	12.50	12.50

Appendix 4 Performance analysis of each of the twenty proposed variants of GHA-DDR in comparison with optimal TE/L based on RPD

Problem instances	Optimal TE/L	RPD of each of the proposed variants of GHA-DDR in comparison with optimal TE/L									
		GHA-DDR1	GHA-DDR2	GHA-DDR3	GHA-DDR4	GHA-DDR5	GHA-DDR6	GHA-DDR7	GHA-DDR8	GHA-DDR9	GHA-DDR10
1	40	55.00	55.00	55.00	55.00	55.00	55.00	55.00	65.00	65.00	55.00
2	60	30.00	30.00	30.00	30.00	30.00	30.00	30.00	36.67	36.67	30.00
3	71	39.44	39.44	39.44	39.44	39.44	39.44	39.44	22.54	22.54	39.44
4	91	17.58	17.58	17.58	17.58	17.58	17.58	17.58	26.37	26.37	17.58
5	135	11.85	22.22	22.22	22.22	11.85	11.85	25.19	22.22	17.78	11.85
6	102	41.18	41.18	41.18	41.18	41.18	41.18	41.18	66.67	66.67	41.18
7	73	106.85	106.85	106.85	106.85	106.85	106.85	106.85	52.05	52.05	106.85
8	80	80.00	80.00	80.00	80.00	80.00	80.00	80.00	87.50	87.50	80.00
9	54	148.15	166.67	166.67	166.67	148.15	148.15	248.15	148.15	148.15	148.15
10	59	57.63	57.63	57.63	57.63	57.63	57.63	57.63	57.63	57.63	57.63
11	68	185.29	185.29	185.29	185.29	185.29	185.29	185.29	185.29	185.29	185.29
12	116	51.72	51.72	51.72	51.72	51.72	51.72	87.93	101.72	101.72	51.72
13	84	66.67	66.67	66.67	66.67	66.67	66.67	66.67	78.57	78.57	66.67
14	52	151.92	151.92	151.92	151.92	151.92	151.92	151.92	151.92	151.92	151.92
15	37	297.30	297.30	297.30	297.30	297.30	297.30	297.30	297.30	297.30	297.30
Problem instances	RPD of each of the proposed variants of GHA-DDR in comparison with optimal TE/L										
	GHA-DDR11	GHA-DDR12	GHA-DDR13	GHA-DDR14	GHA-DDR15	GHA-DDR16	GHA-DDR17	GHA-DDR18	GHA-DDR19	GHA-DDR20	
1	55.00	55.00	55.00	20.00	50.00	50.00	50.00	50.00	50.00	100.00	
2	30.00	30.00	30.00	33.33	25.00	21.67	21.67	21.67	21.67	21.67	
3	39.44	39.44	39.44	16.90	19.72	16.90	16.90	16.90	16.90	33.80	
4	17.58	17.58	17.58	21.98	16.48	20.88	20.88	20.88	20.88	47.25	
5	11.85	11.85	11.85	11.85	4.44	4.44	4.44	4.44	4.44	22.22	
6	41.18	41.18	41.18	7.84	21.57	21.57	21.57	21.57	21.57	41.18	
7	106.85	106.85	106.85	52.05	63.01	63.01	63.01	63.01	63.01	63.01	
8	80.00	80.00	80.00	0.00	62.50	62.50	62.50	62.50	62.50	62.50	
9	148.15	148.15	148.15	118.52	122.22	159.26	122.22	159.26	159.26	111.11	
10	57.63	57.63	57.63	57.63	57.63	57.63	57.63	57.63	57.63	84.75	
11	185.29	185.29	185.29	61.76	79.41	79.41	138.24	79.41	79.41	111.76	
12	51.72	51.72	51.72	24.14	24.14	24.14	24.14	24.14	24.14	24.14	
13	66.67	66.67	66.67	59.52	57.14	57.14	57.14	57.14	57.14	57.14	
14	151.92	151.92	151.92	75.00	75.00	75.00	75.00	75.00	75.00	75.00	
15	297.30	297.30	297.30	21.62	21.62	21.62	21.62	21.62	21.62	21.62	

Appendix 5: Performance analysis of each of the twenty proposed variants of GHA-DDR in comparison with optimal Lmax based on RPD

Problem instances	Optimal Lmax	RPD of each of the proposed variants of GHA-DDR in comparison with optimal Lmax									
		GHA-DDR1	GHA-DDR2	GHA-DDR3	GHA-DDR4	GHA-DDR5	GHA-DDR6	GHA-DDR7	GHA-DDR8	GHA-DDR9	GHA-DDR10
1	21	23.81	23.81	23.81	23.81	23.81	23.81	23.81	0.00	0.00	23.81
2	21	23.81	23.81	23.81	23.81	23.81	23.81	23.81	0.00	0.00	23.81
3	21	23.81	23.81	23.81	23.81	23.81	23.81	23.81	0.00	0.00	23.81
4	26	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7.69	7.69	0.00
5	37	13.51	13.51	13.51	13.51	13.51	13.51	13.51	35.14	0.00	13.51
6	20	30.00	30.00	30.00	30.00	30.00	30.00	30.00	105.00	105.00	30.00
7	24	16.67	16.67	16.67	16.67	16.67	16.67	16.67	16.67	16.67	16.67
8	25	20.00	20.00	20.00	20.00	20.00	20.00	20.00	56.00	56.00	20.00
9	14	7.14	7.14	7.14	7.14	7.14	7.14	42.86	7.14	7.14	7.14
10	16	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25
11	13	23.08	23.08	23.08	23.08	23.08	23.08	23.08	23.08	23.08	23.08
12	30	30.00	30.00	30.00	30.00	30.00	30.00	73.33	83.33	83.33	30.00
13	8	62.50	62.50	62.50	62.50	62.50	62.50	62.50	187.50	187.50	62.50
14	15	20.00	20.00	20.00	20.00	20.00	20.00	20.00	20.00	20.00	20.00
15	9	22.22	22.22	22.22	22.22	22.22	22.22	22.22	22.22	22.22	22.22

Problem instances	RPD of each of the proposed variants of GHA-DDR in comparison with optimal Lmax									
	GHA-DDR11	GHA-DDR12	GHA-DDR13	GHA-DDR14	GHA-DDR15	GHA-DDR16	GHA-DDR17	GHA-DDR18	GHA-DDR19	GHA-DDR20
1	23.81	23.81	23.81	23.81	28.57	28.57	28.57	28.57	28.57	71.43
2	23.81	23.81	23.81	0.00	28.57	28.57	28.57	28.57	28.57	28.57
3	23.81	23.81	23.81	0.00	4.76	4.76	4.76	4.76	4.76	28.57
4	0.00	0.00	0.00	7.69	3.85	3.85	3.85	3.85	3.85	46.15
5	13.51	13.51	13.51	13.51	2.70	2.70	2.70	2.70	2.70	2.70
6	30.00	30.00	30.00	15.00	80.00	80.00	80.00	80.00	80.00	80.00
7	16.67	16.67	16.67	16.67	16.67	16.67	16.67	16.67	16.67	16.67
8	20.00	20.00	20.00	20.00	60.00	60.00	60.00	60.00	60.00	60.00
9	7.14	7.14	7.14	271.43	7.14	50.00	7.14	50.00	50.00	157.14
10	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	6.25	68.75
11	23.08	23.08	23.08	130.77	15.38	15.38	53.85	15.38	15.38	38.46
12	30.00	30.00	30.00	36.67	36.67	36.67	36.67	36.67	36.67	36.67
13	62.50	62.50	62.50	400.00	150.00	150.00	150.00	150.00	150.00	150.00
14	20.00	20.00	20.00	20.00	20.00	20.00	20.00	20.00	20.00	20.00
15	22.22	22.22	22.22	11.11	11.11	11.11	11.11	11.11	11.11	11.11

Appendix 6: Performance analysis of each of the twenty proposed variants of GHA-DDR in comparison with estimated optimal TWT based on ARPD score

S. No.	Problem configuration	ARPD score of each of the proposed variants of GHA-DDR in comparison with estimated optimal TWT									
		GHA-DDR1	GHA-DDR2	GHA-DDR3	GHA-DDR4	GHA-DDR5	GHA-DDR6	GHA-DDR7	GHA-DDR8	GHA-DDR9	GHA-DDR10
1	J1.A1, D1	53.12	54.65	54.65	54.65	53.12	53.79	44.38	55.86	63.17	53.79
2	J1.A1, D2	59.46	82.96	82.96	82.96	59.46	58.71	66.49	92.89	103.81	58.71
3	J1, A1, D3	152.71	219.08	219.08	219.08	152.71	152.71	164.72	150.20	155.90	152.71
4	J1, A2, D1	89.49	94.30	94.30	94.30	89.49	90.30	82.00	144.90	152.33	90.30
5	J1, A2, D2	247.90	220.21	220.21	220.21	247.90	247.09	128.69	221.13	278.96	247.09
6	J1, A2, D3	105.09	180.45	180.45	180.45	105.09	105.09	127.11	120.86	134.45	105.09
7	J1, A3, D1	57.56	56.08	56.08	56.08	57.56	57.56	46.00	155.74	156.72	57.56
8	J1, A3, D2	90.87	117.22	117.22	117.22	90.87	93.70	97.60	142.79	181.55	93.70
9	J1, A3, D3	120.41	173.58	173.58	173.58	120.41	120.41	165.21	246.28	280.79	120.41
10	J2.A1, D1	118.03	111.15	111.15	111.15	118.03	109.42	78.65	124.10	126.07	109.42
11	J2.A1, D2	74.90	120.01	120.01	120.01	74.90	73.17	88.36	100.55	109.14	73.17
12	J2, A1, D3	91.65	173.39	173.39	173.39	91.65	92.94	121.19	115.31	129.06	92.94
13	J2, A2, D1	134.21	138.21	138.21	138.21	134.21	138.44	107.61	130.43	131.02	138.44
14	J2, A2, D2	155.44	163.02	163.02	163.02	155.44	149.74	122.05	191.10	205.17	149.74
15	J2, A2, D3	185.21	229.86	229.86	229.86	185.21	180.35	177.63	211.60	226.70	180.35
16	J2, A3, D1	140.55	141.87	141.87	141.87	140.55	141.45	109.27	194.19	194.97	141.45
17	J2, A3, D2	157.98	183.43	183.43	183.43	157.98	182.57	122.41	183.93	188.70	182.57
18	J2, A3, D3	196.67	234.83	234.83	234.83	196.67	198.48	171.85	230.16	238.08	198.48
19	J3, A1, D1	75.81	80.04	80.04	80.04	75.81	79.39	41.07	66.93	68.46	79.39
20	J3, A1, D2	93.98	90.45	90.45	90.45	93.98	90.71	43.75	70.63	74.39	90.61

S. No.	Problem configuration	ARPD score of each of the proposed variants of GHA-DDR in comparison with estimated optimal TWT									
		GHA-DDR1	GHA-DDR2	GHA-DDR3	GHA-DDR4	GHA-DDR5	GHA-DDR6	GHA-DDR7	GHA-DDR8	GHA-DDR9	GHA-DDR10
21	J3, A1, D3	98.03	110.72	110.72	110.72	98.03	92.42	61.02	72.44	82.17	92.42
22	J3, A2, D1	106.93	106.88	106.88	106.88	106.93	104.62	51.46	80.49	80.69	104.62
23	J3, A2, D2	97.54	109.79	109.79	109.79	97.54	107.39	56.90	82.11	83.97	107.39
24	J3, A2, D3	129.11	144.65	144.65	144.65	129.11	124.09	86.72	107.12	105.67	124.09
25	J3, A3, D1	99.34	107.88	107.88	107.88	99.34	105.02	62.84	91.01	92.36	105.02
26	J3, A3, D2	118.92	120.47	120.47	120.47	118.92	122.00	65.65	93.02	97.02	122.00
27	J3, A3, D3	133.23	132.27	132.27	132.27	133.23	124.85	76.90	94.22	104.62	124.85
S. No.	ARPD score of each of the proposed variants of GHA-DDR in comparison with estimated optimal TWT										
	GHA-DDR11	GHA-DDR12	GHA-DDR13	GHA-DDR14	GHA-DDR15	GHA-DDR16	GHA-DDR17	GHA-DDR18	GHA-DDR19	GHA-DDR20	
1	53.12	53.79	53.12	165.02	0.26	3.70	4.38	2.06	2.06	2.55	
2	59.46	58.71	59.46	268.97	9.54	4.24	6.50	3.86	3.93	20.82	
3	152.71	152.71	152.71	765.51	3.14	47.60	47.41	46.43	46.43	65.32	
4	89.49	90.30	89.49	182.12	5.59	2.53	2.62	1.78	1.78	5.80	
5	247.90	247.09	247.90	533.80	37.76	1.22	20.56	0.59	0.89	33.53	
6	105.09	105.09	105.09	499.80	18.01	5.44	12.19	1.52	1.52	27.67	
7	57.56	57.56	57.56	199.47	3.69	6.29	8.34	5.43	4.92	8.93	
8	90.87	93.70	90.87	172.65	8.84	2.83	8.12	2.03	2.03	18.28	
9	120.41	120.41	120.41	496.82	6.61	4.53	10.99	4.57	4.57	24.85	
10	118.03	109.55	118.03	109.22	2.26	3.74	2.68	2.20	1.89	6.27	
11	74.90	73.17	74.90	101.96	6.67	5.52	7.21	2.69	1.40	16.32	
12	91.65	92.94	91.65	145.22	8.22	10.42	11.96	9.55	6.02	2.20	
13	134.21	138.44	134.21	127.14	3.11	6.82	5.03	1.04	1.28	11.71	
14	155.44	149.74	155.44	152.84	22.60	8.88	18.53	2.94	1.31	23.31	
15	185.21	180.35	185.21	202.49	18.10	8.57	21.11	4.65	1.50	16.25	
16	140.55	141.45	140.55	183.36	19.77	9.65	11.66	0.57	0.93	15.53	
17	157.98	182.57	157.98	165.69	24.65	7.23	20.92	2.59	0.85	47.37	
18	196.67	198.48	196.67	211.44	47.41	6.39	39.17	3.80	3.36	47.04	
19	75.81	80.31	75.81	58.08	3.94	1.61	3.56	0.13	1.00	7.87	
20	93.98	90.64	93.98	60.76	1.68	4.25	2.24	0.23	2.62	9.16	
21	98.03	95.31	98.03	58.15	0.99	4.31	2.91	1.37	1.22	9.53	
22	106.93	103.67	106.93	74.43	4.19	2.73	0.26	0.13	1.37	6.12	
23	97.54	107.39	97.54	74.04	5.57	6.92	4.41	0.87	0.96	12.16	
24	129.11	124.09	129.11	75.75	10.54	7.26	11.89	0.28	1.85	28.51	

S. No. ARPD score of each of the proposed variants of GHA-DDR in comparison with estimated optimal TWT

	GHA-DDR11	GHA-DDR12	GHA-DDR13	GHA-DDR14	GHA-DDR15	GHA-DDR16	GHA-DDR17	GHA-DDR18	GHA-DDR19	GHA-DDR20
25	99.34	104.39	99.34	76.33	3.26	4.85	1.98	0.77	1.43	6.15
26	118.92	122.00	118.92	90.27	5.90	7.24	3.05	0.34	0.82	19.52
27	133.23	124.85	133.23	89.45	17.01	14.11	6.76	1.40	2.85	18.04

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